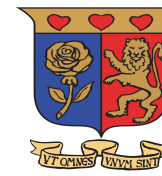


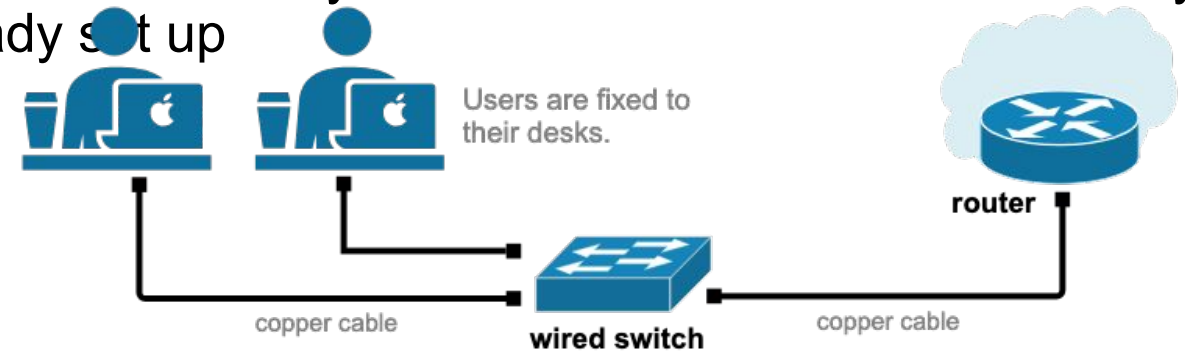
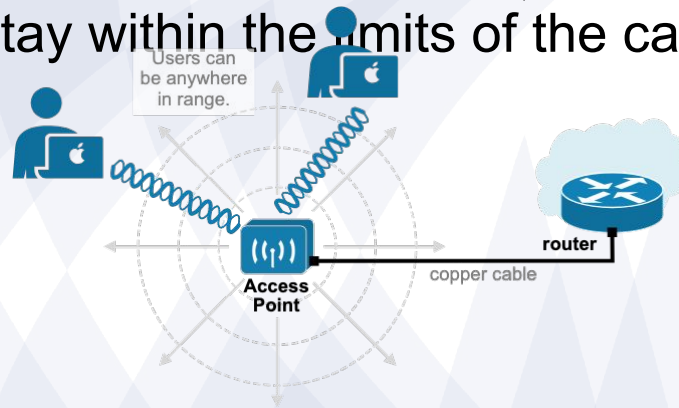
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Wireless Networks and WLANs

Wireless Vs Wired LANs

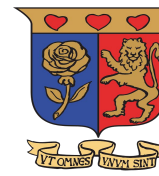


- The most obvious property of wired networks is that a wire must physically connect any two devices that want to communicate. Additionally, devices do not connect with ANY kind of wires. Physical cables used in networking follow strict standards (IEEE802.3) for length, quality, foil, connectors, fire resistance, and so on. Therefore, any two devices are connected via a standardized physical cable.
- A wired connection is fixed, meaning users cannot easily move their devices around. They must stay within the limits of the cables already set up



- A wireless network eliminates the need for wires. It uses radio frequency signals to connect devices, allowing users to move freely while staying connected. Additionally, it allows users to easily connect multiple wireless devices to the network as there's no limitation due to the number of cables or ports
- However, wireless data travels through open space, as illustrated in the diagram above. Wireless data lacks this security unlike a wired connection, which provides a protected, dedicated link between two devices. This open environment brings various factors that can affect data transmission and delivery.

Wireless Vs Wired LANs

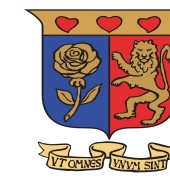


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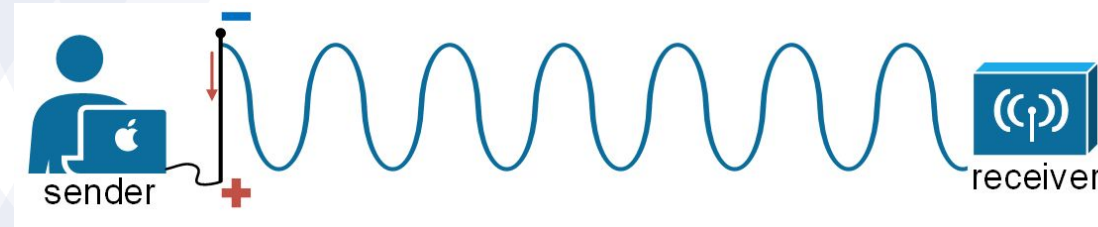
- It's clear that people are willing to trade speed, reliability, and security for the convenience of mobility and roaming.
- This need for mobility is precisely why wireless networks were created: to use mobile devices everywhere. To connect at the local cafe. To connect at the office. To connect at home. Mobility outweighs all other disadvantages.
- Network engineers are responsible for ensuring that mobility and roaming work flawlessly in organizations' wireless network because this is what wireless is all about.
- They must however, offset the disadvantages of wireless technology.

	Wired Networks	Wireless Networks
Medium	Cables (copper, fiber).	Radio waves.
Speed	Faster (can reach up to 400Gbps)	Slower than wired networks (typically up to 1-6 Gbps in perfect conditions).
Latency	Lower.	Higher latency than wired networks. Prone to device congestion.
Reliability	More reliable. Less prone to interference.	Less reliable. Prone to signal interference.
Mobility	Limited to cable installation and length.	High mobility, supports roaming.
Installation	Requires cabling and physical setup.	Easier, no cables required.
Security	More secure, physical access needed.	Less secure, vulnerable to sniffing.

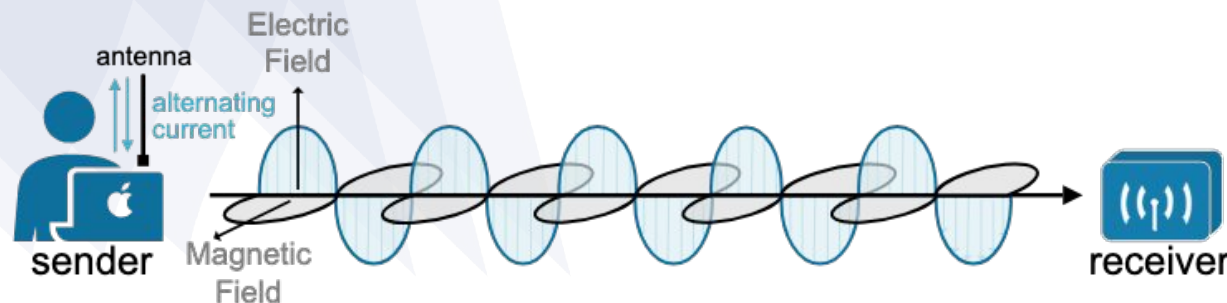
Wireless LANs: Transmission Fundamentals



- In a wired network, data is converted to an electrical signal and applied to one end of a cable. The cable is conductive, so the signal travels to the other end at the speed of light. The same logic applies when data is transmitted over fiber cables.
- In contrast, a wireless connection doesn't use physical wires to carry the signal; instead, it uses radio waves to transmit data over free space. A device's wifi transmitter sends an alternating current through the antenna. The current is changing directions (continuously changing between a positive and negative voltage) resulting in a waveform

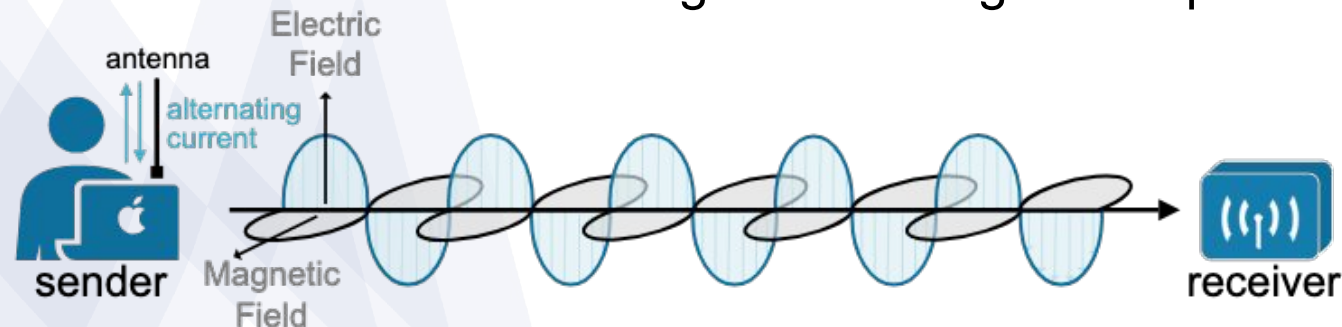


- The antenna also creates a magnetic field that is perpendicular to the electric field. These fields move together, always at right angles to each other, and spread outward in all directions.

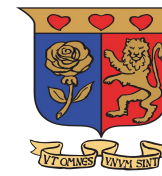


Wireless LANs: Transmission Fundamentals

- Notice that the antenna is vertically aligned. The electric field is aligned with the orientation of the antenna. The magnetic field is always perpendicular to the electric field. In this case, the magnetic field is horizontal. This alignment is why the polarization of the transmitted wave (which depends on the electric field's orientation) is vertical for a vertically aligned antenna.
- At the receiver's end, the antenna picks up these electromagnetic waves. As the waves hit the antenna, they create an electrical signal. If everything works properly, the receiver gets a signal similar to the one sent by the transmitter.
- This is the foundation of electromagnetism and how wireless communication works—turning electric signals into radio waves and sending them through free space to a receiver.



Wireless LANs: Frequency



- Radio waves have multiple properties, the most important of which is frequency. The frequency is the number of times the wave completes a full up-and-down cycle in one second. It is measured in Hertz (Hz). The following diagram shows how to measure and calculate the frequency of an electromagnetic wave.

Frequency (F) is the number of completed cycles per second, measured in Hertz (Hz).

$$F = n / t$$

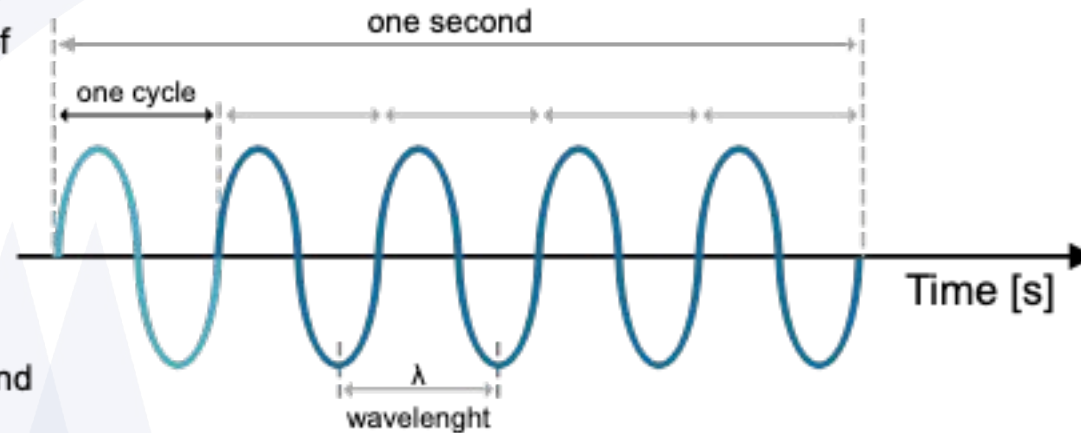
F – frequency

n – number of cycles

t – time in seconds

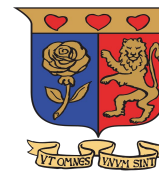
Frequency = 5 cycles / 1 second

F = 5 Hertz (Hz)



- A cycle starts when the wave rises from the center line, falls below it, and rises back to the center line. In the example, the wave makes five cycles in one second. Hence, its frequency is 5 hertz (Hz). However, in nature, there are very high frequencies, ranging from 1 Hz to 10^{23} Hertz (yes, 1 with 23 zeros!). This is currently the upper limit of frequencies observed in nature, as higher frequencies would require incredibly high-energy processes not yet observed.

Wireless LANs: Wavelength



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- RF signals are often described by their frequency, but it can be hard to imagine their size as they move through space. Wavelength measures the physical distance a wave travels during one complete cycle. The Greek symbol for wavelength is lambda (λ).

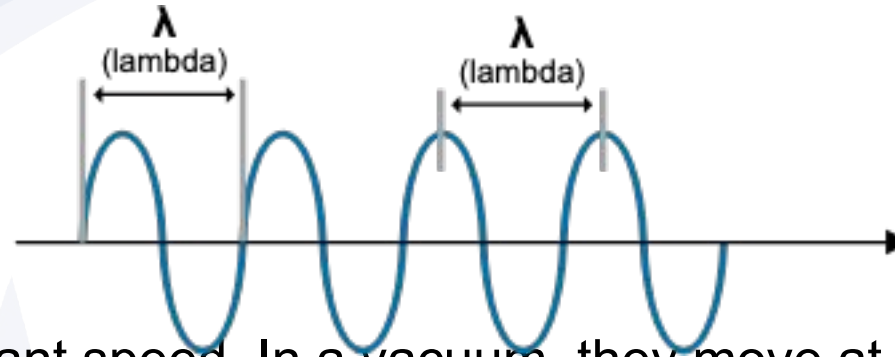
Wavelength (λ) is the distance a wave travels during one complete cycle.

$$\lambda = c / f$$

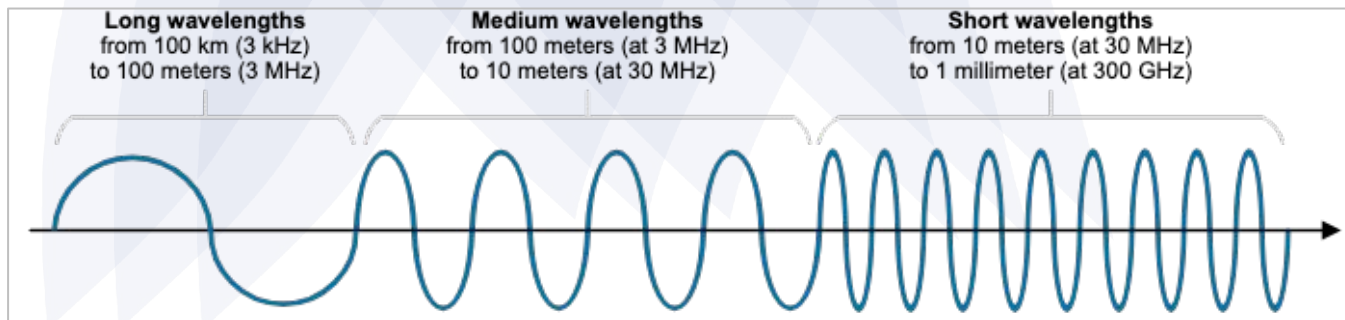
λ is the wavelength in meters.

c is the speed of light in a vacuum.

f is the frequency in hertz (Hz).



- RF waves always travel at a constant speed. In a vacuum, they move at the speed of light, but in air, they are slightly slower. Notice that as the frequency increases, the wavelength gets shorter. Higher frequencies mean smaller waves that cover less distance. To remember the relation between frequency and wavelength - you can try to calculate the wavelength of given frequencies. For example, calculate the wavelength of 6GHz.



$$\lambda = c / f$$

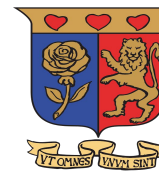
where λ is the wavelength in meters,

c is the speed of light in a vacuum (3×10^8 meters),

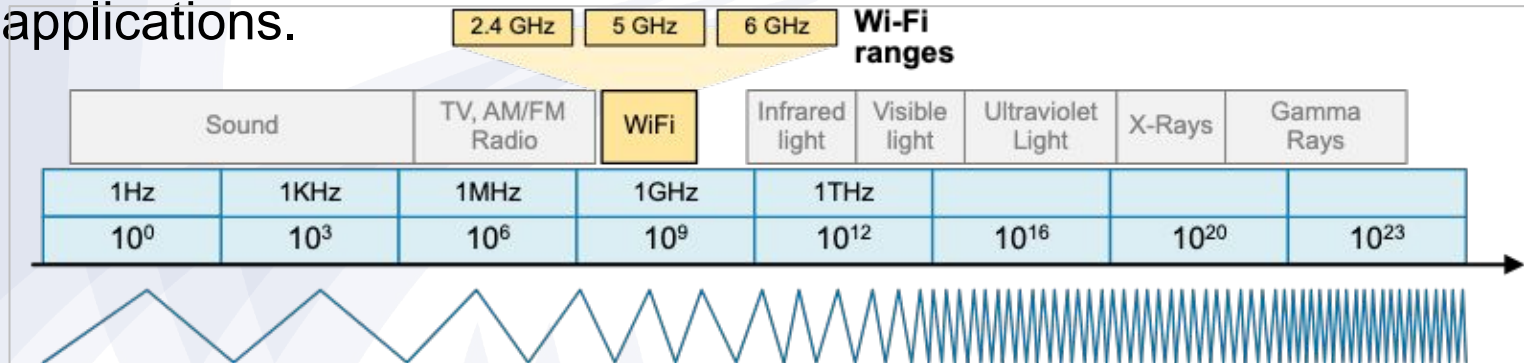
f is the frequency in hertz (6×10^9 Hz).

$$\lambda = 3 \times 10^8 / 6 \times 10^9 = 0.05 \text{ meters} = 5 \text{ cm}$$

Wireless LANs: Bands and Channels

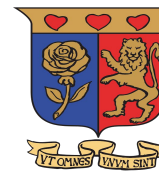


- Now that we know what frequency is. in nature, there is a very wide range of frequencies (from 1 Hz to 1023 Hz); this is the frequency spectrum. The diagram below visualizes it in a very simplified way. the spectrum is finite, and certain frequency ranges are more suitable for specific applications.



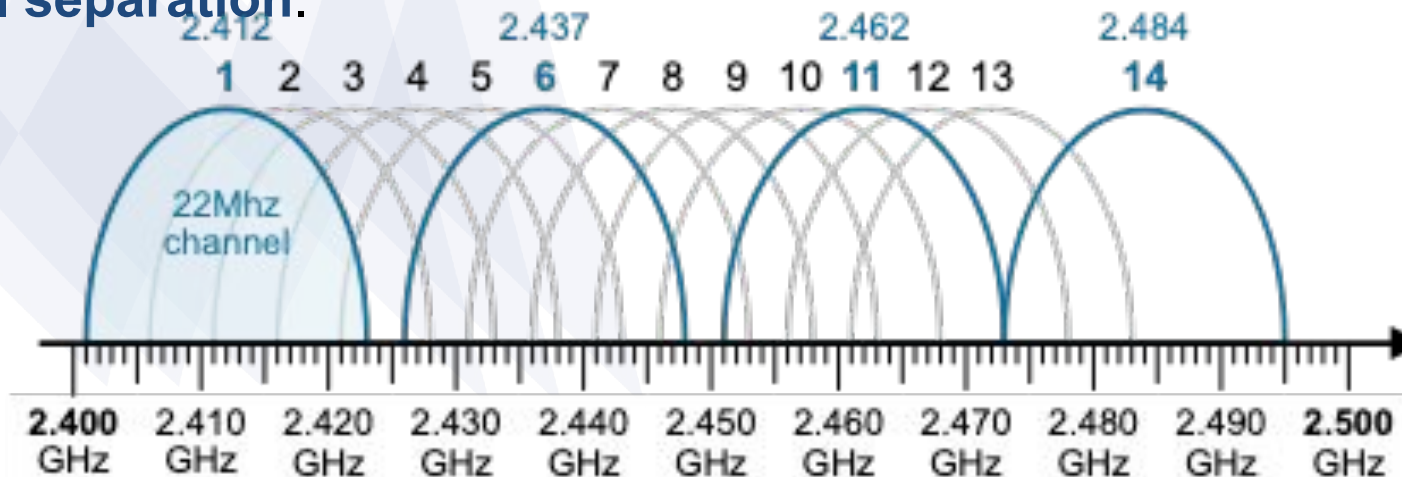
- Frequencies are crucial for defense, intelligence, and public safety systems. That's why governments issue spectrum licenses to organizations, often through auctions or direct allocation, defining specific terms of use.
- Another important aspect of the frequency band is that certain ranges are more suitable for specific applications. For example, the frequencies between 88 MHz and 108 MHz are best suited for FM radio stations. A continuous range of frequencies used for the same purpose is called a frequency band. For example, AM radio stations use frequencies from 530 kHz to about 1710 kHz. Terrestrial TV broadcasting uses a frequency range between 300 MHz and 700 MHz, and so on.

Wireless LANs: Wi-Fi Bands and Channels

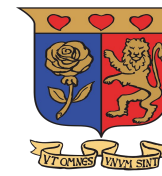


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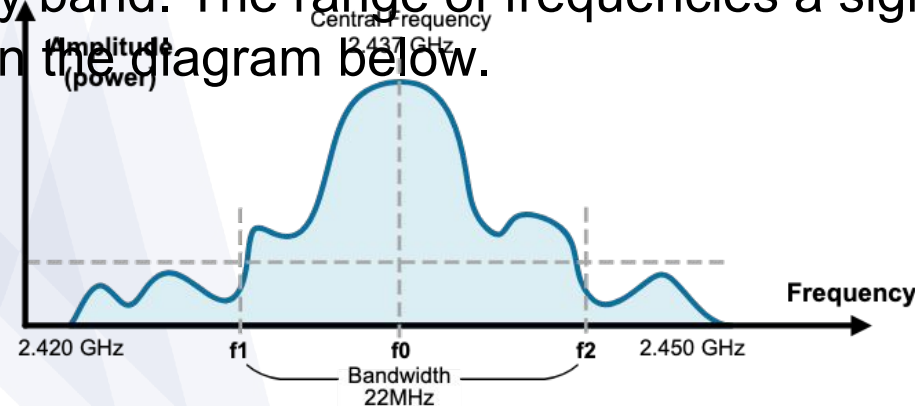
- Wi-Fi has been allocated three bands that are typically the same worldwide (with slight variations in some regions):
 - 2.4 GHz band with a frequency range of 2.4 GHz to 2.4835 GHz.
 - 5 GHz band with a frequency range of 5.15 GHz to 5.825 GHz (which may vary slightly by region).
 - 6 GHz band with a frequency range of 5.925 GHz to 7.125 GHz (which may vary slightly by region).
- Bands are divided into smaller parts called **channels**. Each channel has a number and is assigned a specific frequency. When channels are defined by national or international standards, they can be used the same way everywhere. For example, the following diagram shows the channel of the 2.4-GHz band used for wireless LANs. It has **14 channels**, each with its own specific frequency. It's much easier to refer to channels by their numbers instead of their exact frequencies. The channels are spaced evenly, 0.005 GHz apart. This spacing is called **channel separation**.



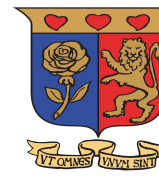
Wireless LANs: Bandwidth



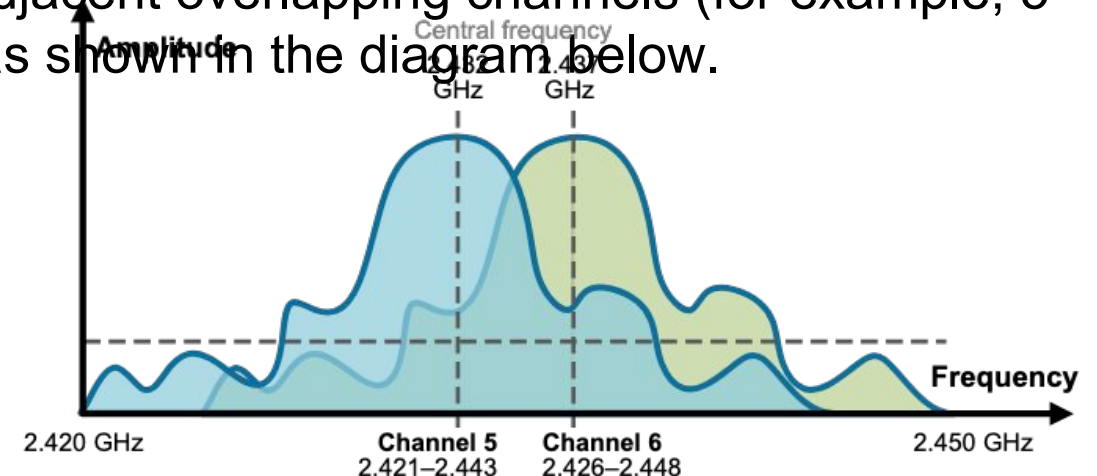
- And you may ask the following question: if the 2.4GHz Wi-Fi band is between 2.400GHz and 2.4835GHz, why don't we have 80 channels? Why do we need 0.005GHz channel spacing and only have 14 channels? Can a device just transmit on a specific frequency like 2.437GHz, another one on 2.438GHz, and another on 2.439GHz, etc.?
- Well, the problem is that RF signals aren't perfectly narrow. They spill over into nearby frequencies, creating a range of frequencies around a central point called the **signal bandwidth**.
- An RF signal isn't perfectly narrow; instead, it spreads slightly above and below its center frequency, taking up nearby frequencies. The center frequency determines where the channel is located within the frequency band. The range of frequencies a signal uses is called its **bandwidth** and is shown in the diagram below.



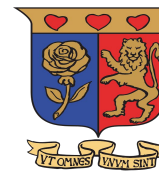
Wireless LANs: Bandwidth



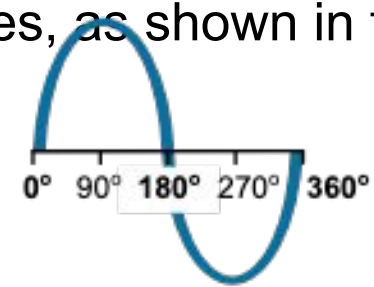
- Frequency bandwidth is the range of frequencies that a specific radio signal occupies. It measures how much "space" the signal takes up on the frequency spectrum. But why does every frequency spread across other frequencies (bandwidth)? Can't we transmit on a single frequency, exactly 2.437 GHz?
- The short answer is yes, we can, but every real-world signal needs some bandwidth to carry information. A signal at an exact frequency (like 2.437 GHz) would be a pure sine wave, and pure sine waves carry no data. To transmit information, we need to modulate the signal (change its amplitude, frequency, or phase), and that spreads the signal across a small range of frequencies around the center frequency.
- For example, if you are transmitting on 2.4GHz Wi-Fi channel 6, the center frequency is 2.437 GHz, but the signal's bandwidth (22 MHz) spreads across nearby frequencies. This is true for all channels. When two wireless devices use adjacent overlapping channels (for example, 5 and 6), their signals interfere with each other, as shown in the diagram below.



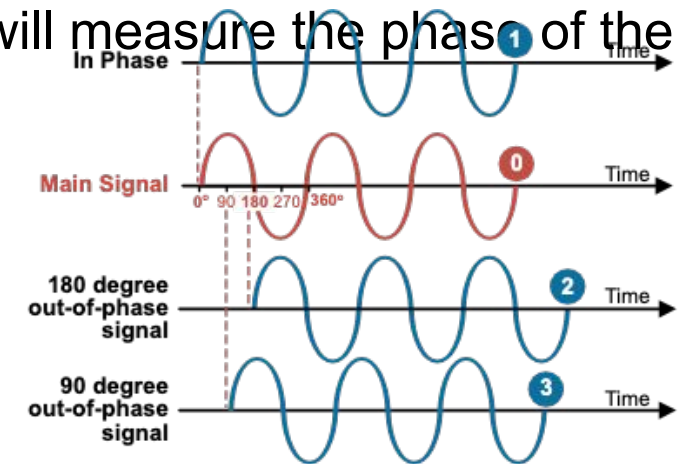
Wireless LANs: Phase



- Phase is another important property of RF signals. However, it isn't a property of a single RF signal; it describes how two or more RF signals with the same frequency relate to each other. Phase describes how the two waveforms' peaks (high points) and troughs (low points) line up. It is typically measured in degrees, where one wave cycle represents 360 degrees. The start of the cycle equals 0 degrees, while the end of the cycle equals 360 degrees, as shown in the diagram below.

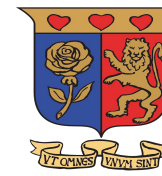


- The following diagram will help you understand the concept. All four waves are RF signals on the same frequency. The wave (0) in red is the main signal. We will measure the phase of the other three waves (1-3) relative to the main signal (0).

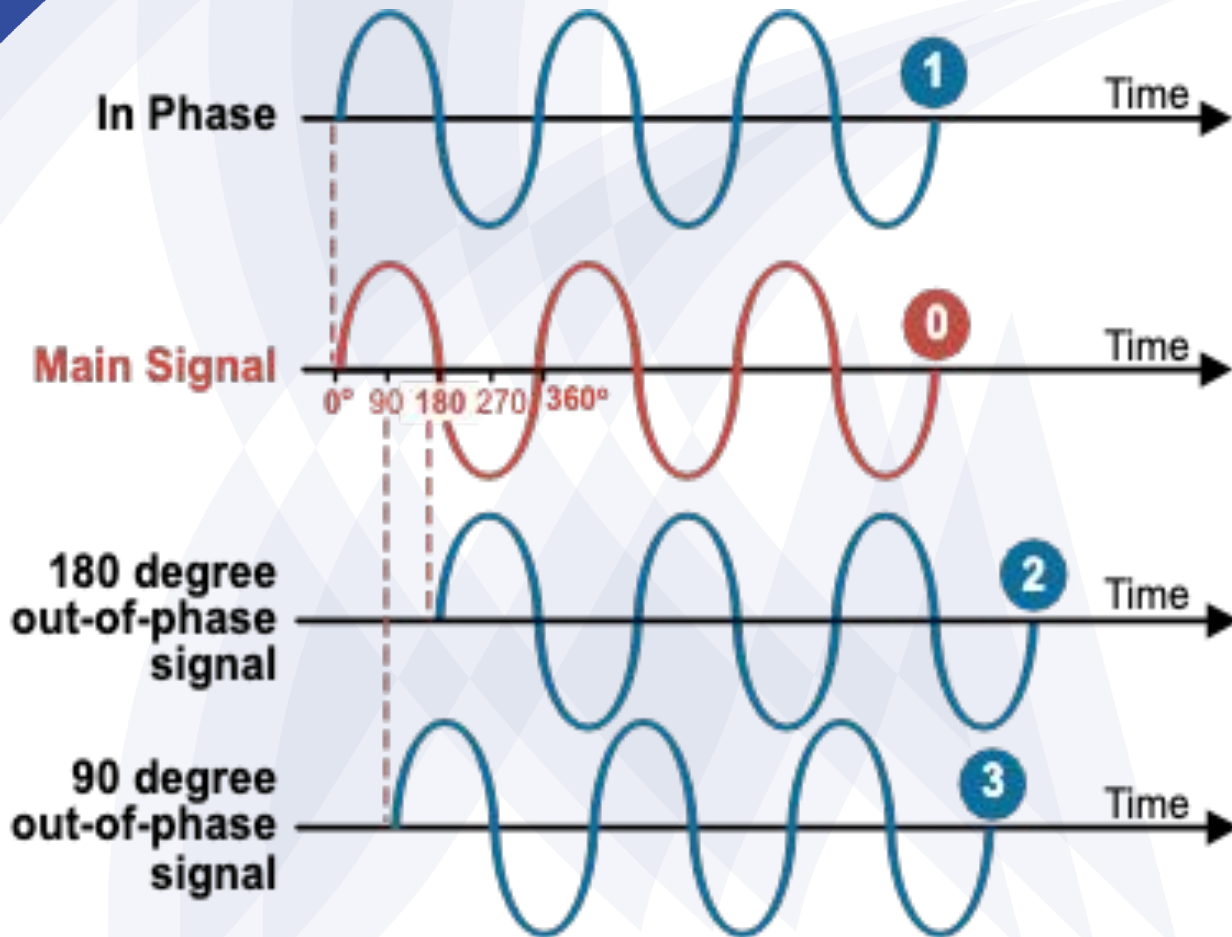


Important to understand: RF signals that are in phase **add together** and strengthen each other. RF signals that are 180 degrees out of phase **cancel each other out**.

Wireless LANs: Phase

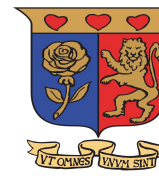


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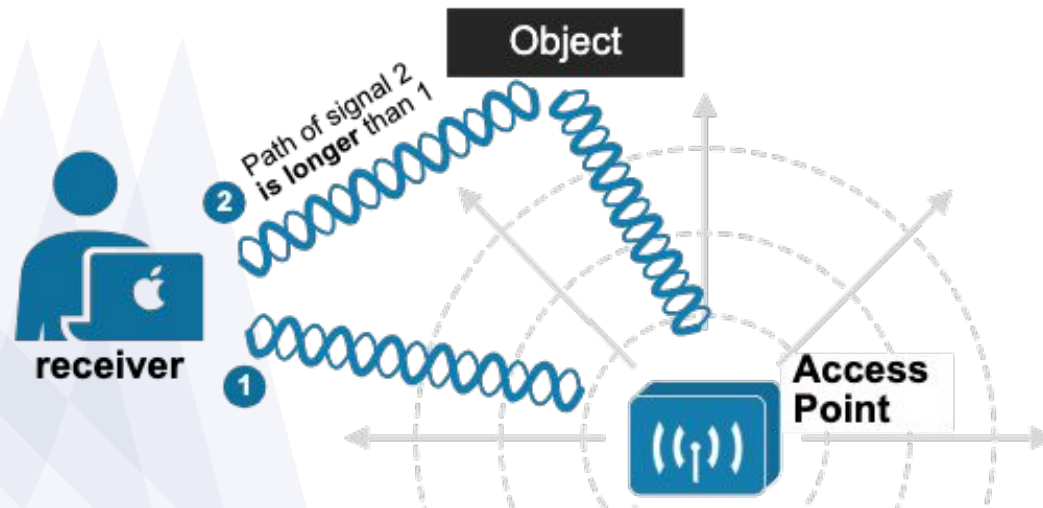
- **Wave 1** - When two signals are in phase, their peaks (high points) and troughs (low points) align perfectly at the same time. This means the two signals are synchronized, and their waveforms match. If represented in degrees, the phase difference between them is 0 degrees.
- **Wave 2** - When two signals are 180 degrees out of phase, their peaks align with the troughs of the other signal, meaning they are completely opposite. When one signal reaches its maximum, the other is at its minimum. This creates a phase difference of 180 degrees, and the two signals effectively cancel each other out if they have the same amplitude.
- **Wave 3** - When two signals are 90 degrees out of phase, one signal is shifted by a quarter of a wavelength compared to the other. This means the peak of one signal aligns with the midpoint (zero crossing) of the other. This creates a phase difference of 90 degrees between the two signals.

Wireless LANs: Whys is Phase Important



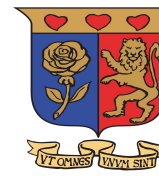
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- At this point, you might be wondering why this is important. When two signals are produced at precisely the same time, they will always be in phase. But are they?
- Two identical RF signals (1 and 2) are produced exactly at the same time by the access point. Signal 1 goes directly to the receiver. However, signal 2 is reflected by an object and takes longer paths than the direct one. Therefore, the receiver gets signal 2 with a delay (due to the longer path). This delay shifts the phase of signal 2 relative to signal 1. If the phase shift happens to be 180 degrees, they can completely cancel each other so the receiver doesn't receive anything.

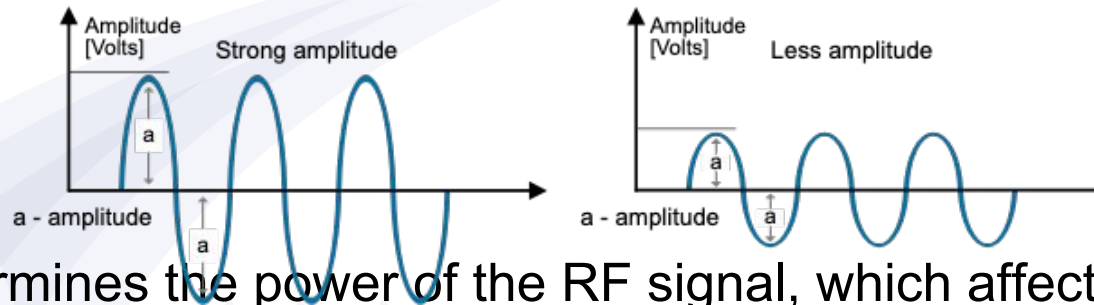


- Depending on how much two signals are out of phase, the strength of the received signal can either increase or decrease. Understanding the phase difference between two signals is key to explaining an RF phenomenon called multipath.

Wireless LANs: Amplitude

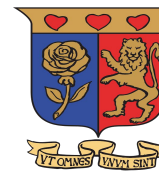


- The amplitude of an RF (Radio Frequency) signal refers to the strength of the oscillating wave that carries the signal. It is essentially the maximum height of the wave from its central baseline (zero point) to its peak (positive or negative), as shown in the diagram below.



- Amplitude determines the power of the RF signal, which affects how far the signal can travel and how well it can be received. A higher amplitude generally means a stronger signal, while a lower amplitude indicates a weaker signal. Amplitude is measured in units like **volts (V)** or **decibels (dB)**, depending on the context. *Wait, but what is a decibel?*
- Decibel (dB) is a logarithmic unit used to measure the strength of a signal. It represents ratios rather than absolute values. A dB value shows how much stronger or weaker one signal is compared to another.
- This makes it useful for efficiently comparing very large or very small values. Although it was initially invented to measure sound levels, people quickly realized that it also applies to power levels. It is a logarithmic scale, which means it compares values using powers of ten instead of simple addition.

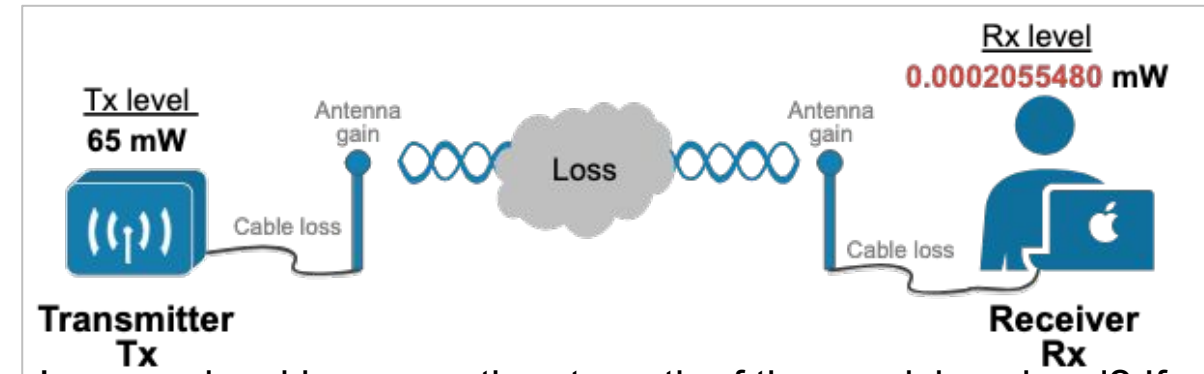
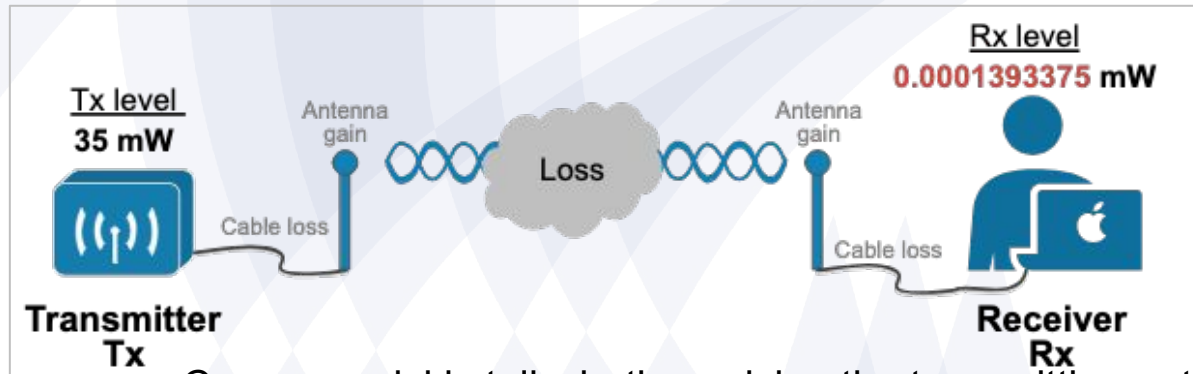
Wireless LANs: Decibels



- In radio communications, engineers must compare very large or very small numbers. Let's consider two imaginary examples that explain why people use decibels instead of absolute values.

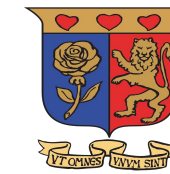
Scenario 1: An antenna transmits with a power level of 35 mW, and a receiver receives a power level of 0.0001393375 mW. But there is a problem. The receiver complains of a bad-quality signal.

Scenario 2: You decide to raise the power level to see if it fixes the problem. The antenna now transmits with a power level of 65 mW, and a receiver receives 0.0002055480 mW.



- Can you quickly tell whether raising the transmitting antenna's power level improves the strength of the receiving signal? If you spend some time calculating, you can, but with decibels, it is obvious instantly whether the receiving signal is better in scenario 1 or 2.
- People realized that working with absolute values is not practical. It is hard for humans to compare large to very small numbers quickly and make sense of them. Let's see the same example with decibels:
 - In scenario 1, the receiving power is degraded by 54dB compared to the transmitting signal.
 - In scenario 2, the receiving power degraded by 55dB compared to the transmitting signal.
- Well, with decibels, it is immediately obvious that in scenario 1, the receiver gets better signals. Lower decibels mean a better signal because the signal degraded less. Decibels represent the ratio between very large and very small values with a simple, human-friendly number.

Wireless LANs: Decibels



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- Let's try to explain how we calculate decibels. You need to understand basic knowledge of logarithms.
- The base-10 logarithm, also known as a common logarithm and denoted as $\log_{10}$, is a way to express how many times a number is multiplied by itself to get a certain number. Let's see a few examples:

$\log_{10}(100)$ is 2 because 10×10 is 100.

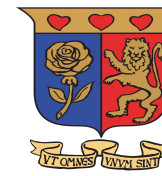
$\log_{10}(100000)$ is 5 because $10 \times 10 \times 10 \times 10 \times 10$ is 100,000.

- The common logarithm is a useful tool in math and science for simplifying large numbers. It is the foundation of decibels. Remember that a decibel is a way of describing the ratio between two values. Therefore, with dB, you will always compare two values. There are two methods to compare two absolute power levels using decibels. The first method uses the following equation:

$dB = 10 \cdot (\log_{10} Power2 - \log_{10} Power1)$ where *Power2* is the receiving power level, and *Power1* is the transmitting power level.

- Power 2 is called the source of interest. We want to understand how much lower the receiving signal is than the original signal. Power 1 is usually called the source of comparison or

Wireless LANs: Decibels



- Let's calculate the example shown in scenario 1

```
#Given power levels in mW
Power1 = 35
Power2 = 0.0001393375

dB = 10.(log10(0.0001393375) - log10(35))
dB = 10.(-1.544068 - 3.855931)
dB = 10.(-5.399999)
dB = -54
```

- There is another simpler equation that can calculate the ratio even more easily. Instead of subtraction, it uses division of the two power levels, as
- Let's use it for scenario 2

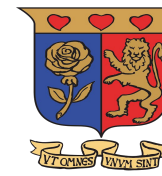
$$dB = 10 \cdot \log_{10}(\text{Power2}/\text{Power1})$$

```
#Given power levels in mW
Power1 = 65
Power2 = 0.0002055480

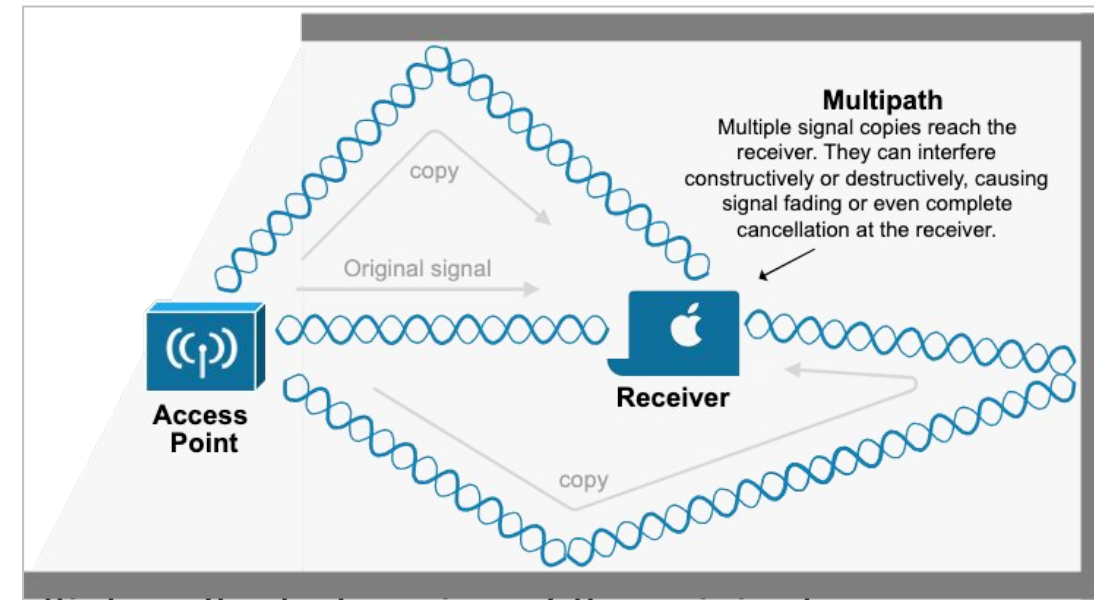
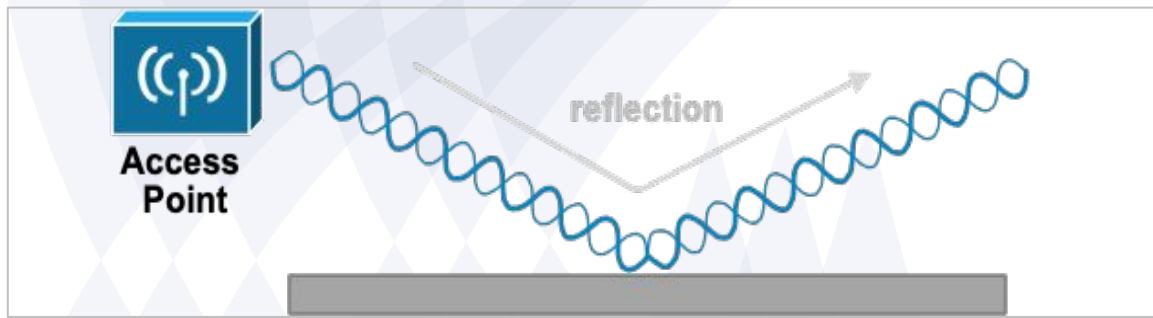
dB = 10.log10(Power2/Power1)
dB = 10.log10(0.0002055480/65)
dB = -55
```

Review dBm, dBi, Isotropic antennae, antennae gain and loss, transmiter and receiver signal strength

Wireless LANs RF : Reflection

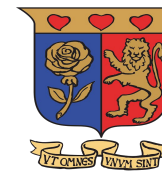


- The visible light is an electromagnetic wave that is part of the electromagnetic spectrum, which includes WiFi bands as well. When light hits a smooth surface, it is reflected at the same angle it hits. All electromagnetic waves can experience reflection. When an RF signal in the Wi-Fi band hits a dense material such as metal or water, it bounces off, as shown in the diagram below.

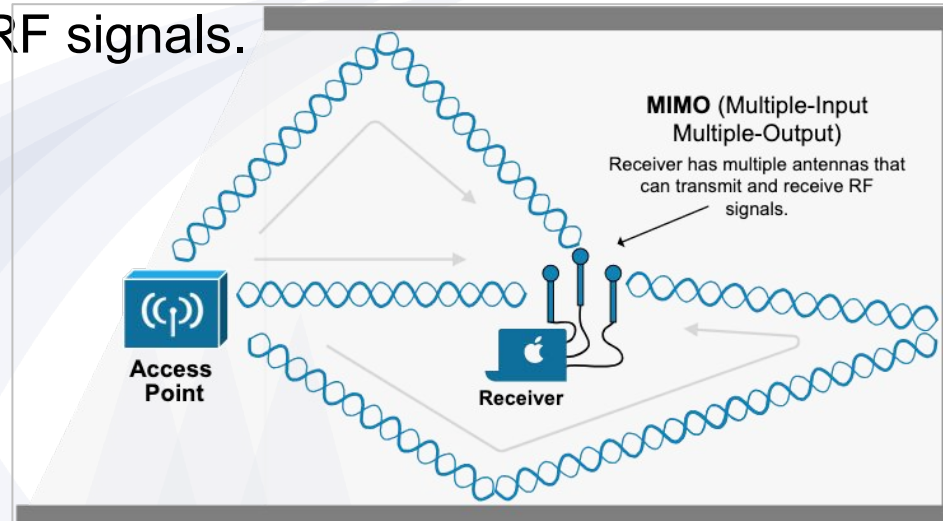


- When there is a reflection, a receiver can receive multiple reflected copies of the original signal. Typically, the copies arrive out of phase because the reflected signal takes a longer path and arrives a bit later, causing what's known as multipath interference. When the receiver gets multiple signals, it mixes them into a weak and distorted version of the original signal, leading to degraded data.

Wireless LANs RF : MIMO

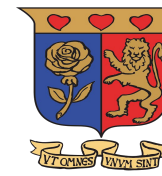


- However, reflection is not always a bad thing. It is possible to take advantage of the fact that the receiver gets multiple signal copies by deploying multiple antennas. A technique called MIMO (Multiple-Input Multiple-Output) means that a device has multiple antennas that can transmit and receive RF signals.



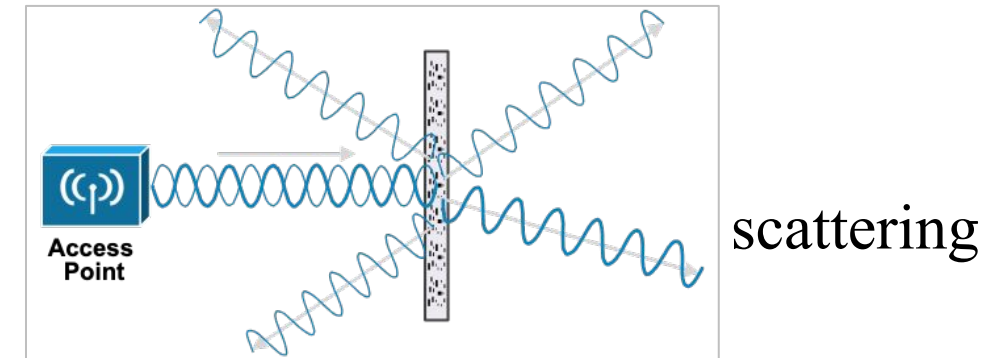
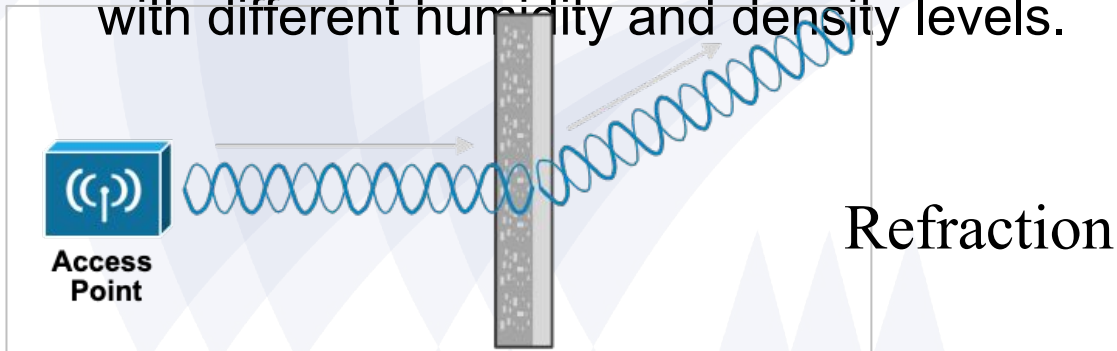
- Imagine WiFi signals bouncing off objects like walls and furniture, creating multiple reflected signals. With just one antenna, these reflections can mix together and cause interference. But with MIMO, multiple antennas are used to receive these different signals. Each antenna captures a separate reflected signal, allowing the WiFi system to process them individually.
- This means it can combine the best parts of each signal, improving the overall data quality and speed. Nowadays, all premium-class wireless devices, laptops, and phones use MIMO. We simply don't see it because the antennas are built-in inside devices (not as shown in the diagram above).

Wireless LANs RF : Refraction and Scattering



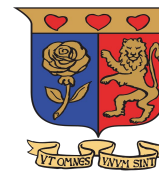
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- Refraction: When an RF signal passes through an object made of materials with different densities, it changes direction (bends), which is called refraction. If we compare it with reflection, reflection is bouncing off a surface, and refraction is bending while passing through a surface. Refraction can also occur when an RF signal passes through layers of air with different humidity and density levels.

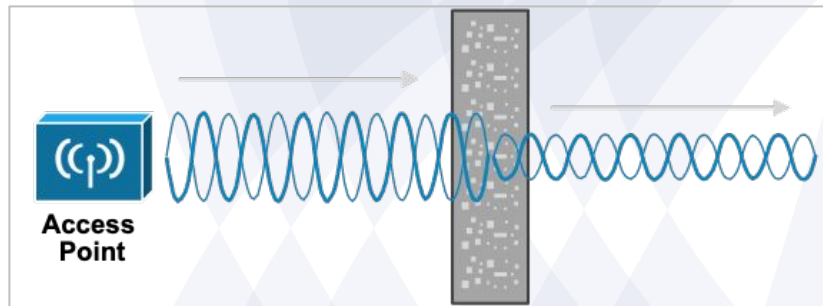


- Scattering: When an RF signal goes through materials made up of different tiny particles, it can scatter in many different directions. Typically, it occurs when a wireless signal travels through a sandy or dusty environment.
- Scattering degrades signal reception. In some cases, it enables signals to reach areas that would otherwise be blocked (non-line-of-sight communication). However, excessive scattering can cause signal distortion and interference, reducing overall wireless performance.
- The effect is more pronounced at higher frequencies (such as 5 GHz WiFi) because shorter wavelengths interact more with small obstacles like dust, rain, or foliage.

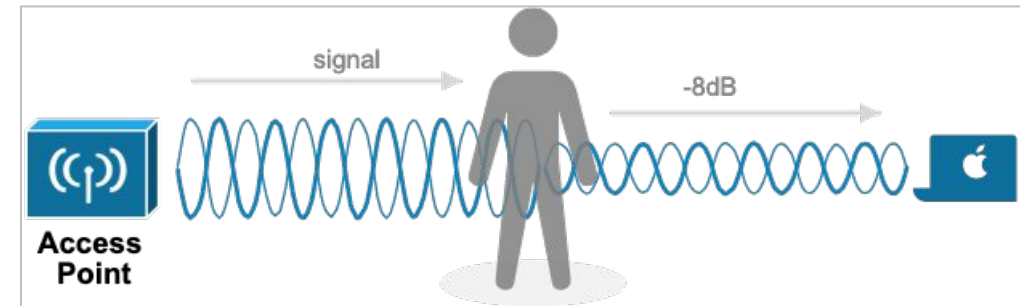
Wireless LANs RF : Absorption



- Absorption: When an RF signal passes through dense materials, it loses energy. The denser the material, the more the signal weakens. Absorption is the most significant factor in indoor signal loss and wireless range limitations. A typical example is when a wireless signal passes through walls in a building or home. Different wall materials absorb different amounts of energy (For instance, drywall might weaken the signal by -4 dBm, while a solid concrete wall might weaken it by -12 dBm). The thicker or denser the wall, the greater the signal attenuation.

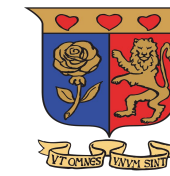


Absorption

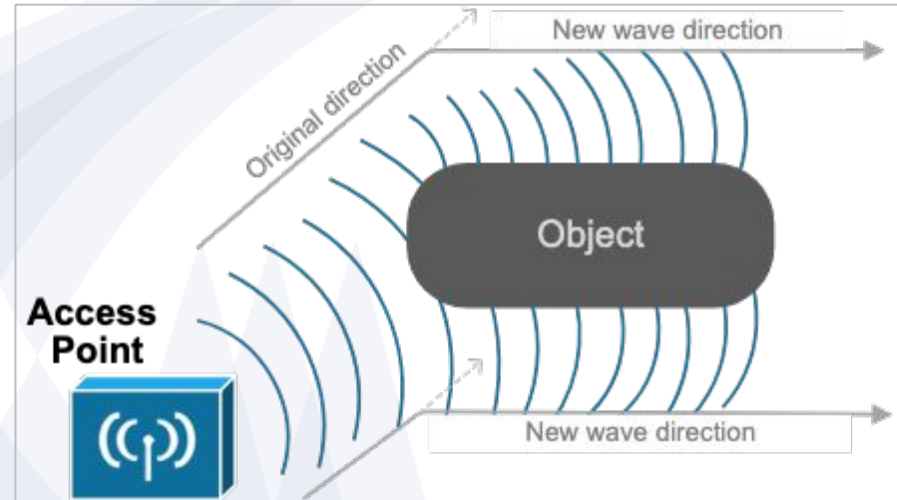


- The human body also absorbs a lot of signals. Depending on how a person is positioned relative to the transmitter, their body might weaken the signal. For example, covering a phone antenna with a hand can decrease the signal by -8 dB, and a person's body might weaken the signal by 15 dB.
- In an office building, each person's body can potentially weaken the signal from a transmitter. That's why wireless access points (APs) are mounted in the ceiling so that the signal does not interfere with furniture, metal objects, and human bodies and gets absorbed

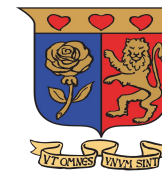
Wireless LANs RF : Diffraction



- Diffraction occurs when an RF signal bends around obstacles or spreads after passing through narrow openings. This allows the signal to reach areas not directly in the access point's line of sight. However, diffraction weakens the signal, reducing its strength and lowering data speeds. It is most noticeable when signals encounter the edges of walls, furniture, or buildings..



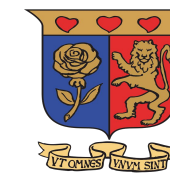
Wireless LANs: Considerations



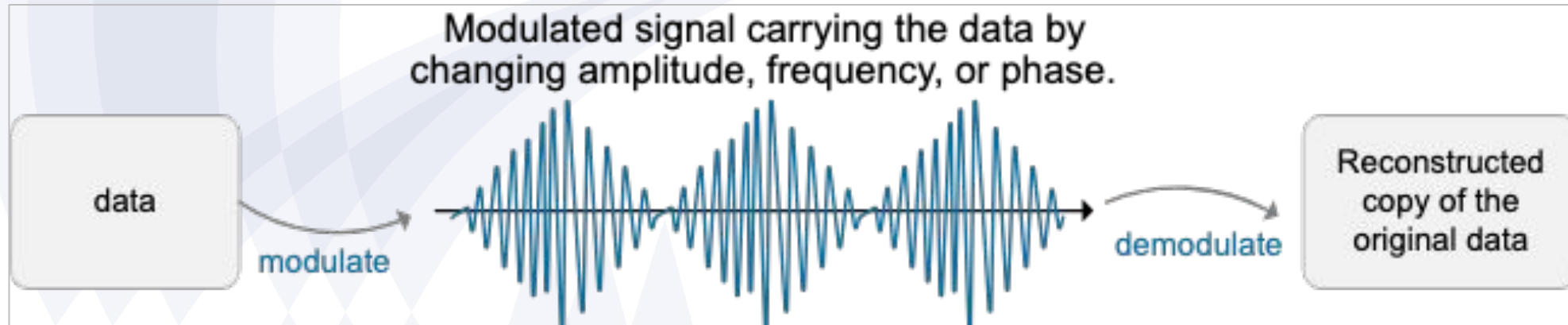
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- When planning for wireless coverage, account for all radio signal effects, such as reflection, refraction, absorption, diffraction, etc. These affect both light and wireless signals are
- Everybody understands how light propagates in free space. When it hits a smooth surface, it reflects. When it goes through translucent material, it loses energy. When it hits dense material, it is scattered or completely absorbed.
- Thinking of wireless signals as light when planning the placement of access points and wireless routers is a good thought process. Of course, some materials are transparent for wi-fi frequencies but are opaque for light. Anyway, the RF effects are very similar, so it is a good starting point.
- Placing wireless access points (APs) in a room, house, or office is similar to positioning light bulbs for even illumination. Just as light spreads from a bulb to cover an area, a Wi-Fi signal radiates from an AP to provide signal coverage. Both light and Wi-Fi signals are forms of electromagnetic waves, meaning they experience similar effects, such as reflection, refraction, absorption, and interference.
- For example, just as shadows form when objects block light, Wi-Fi dead zones appear when walls, furniture, or other obstacles obstruct the signal. Similarly, placing a light bulb too close to a wall may cause uneven lighting, just as an AP placed near obstructions may result in weak or inconsistent connectivity.

Sending Data over RF Signals: Encoding and Modulation

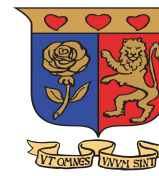


- Modulation is important to send any data over an RF signal. It is when we dynamically change the properties of an RF signal to encode useful information. There are three parts of a signal that we can change: amplitude, frequency, and phase

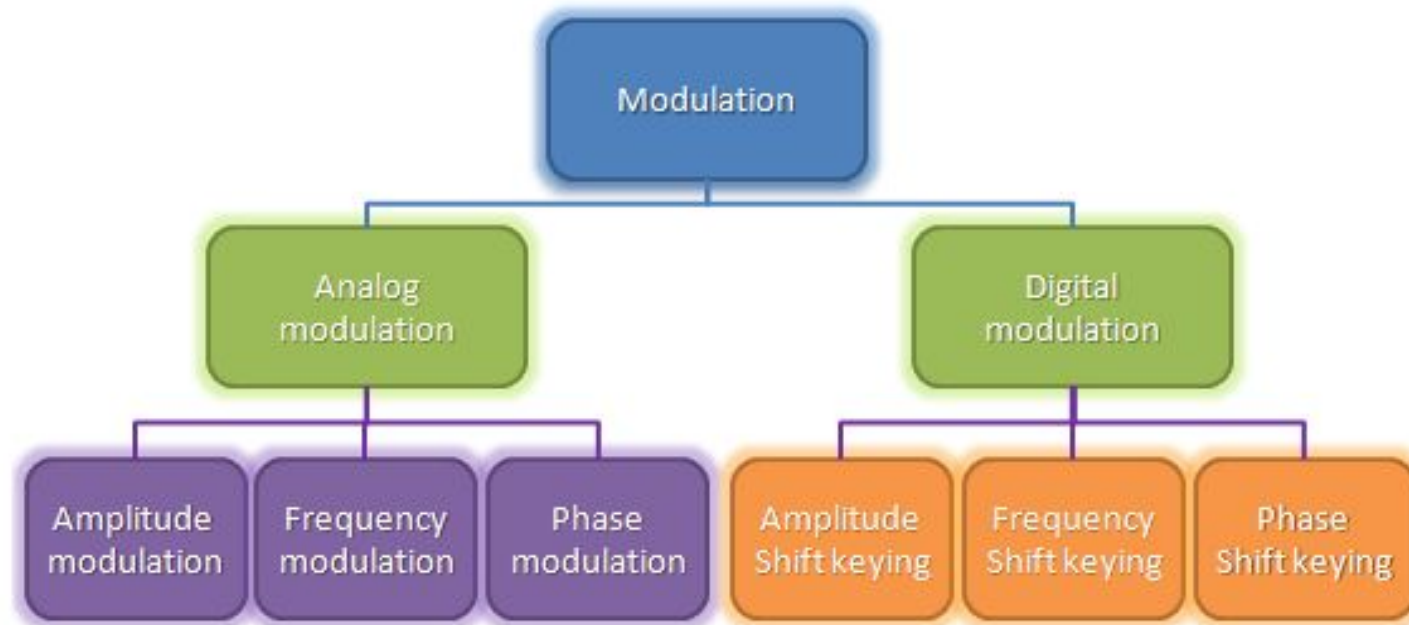


- Analog modulation is a technique for sending a continuous signal with information (most commonly human voice or music) over a higher-frequency RF signal called a carrier. It is called analog because the input signal is not 0s and 1s but an analog signal.
- In analog modulation, the information carried to the remote end is continuous (e.g., voice or video signals). However, most information these days is discrete (binary 0s and 1s). That's why all modern wireless communications (Wi-Fi, LTE, 5G) use digital modulation.

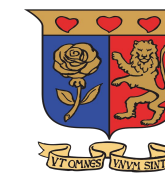
Sending Data over RF Signals: Modulation



- Modulation is the process of varying one or more properties of a periodic waveform, called the carrier signal, with a modulating signal that typically contains information to be transmitted.
- Modulation is the process of encoding information from a message source in a way that is suitable for transmission.

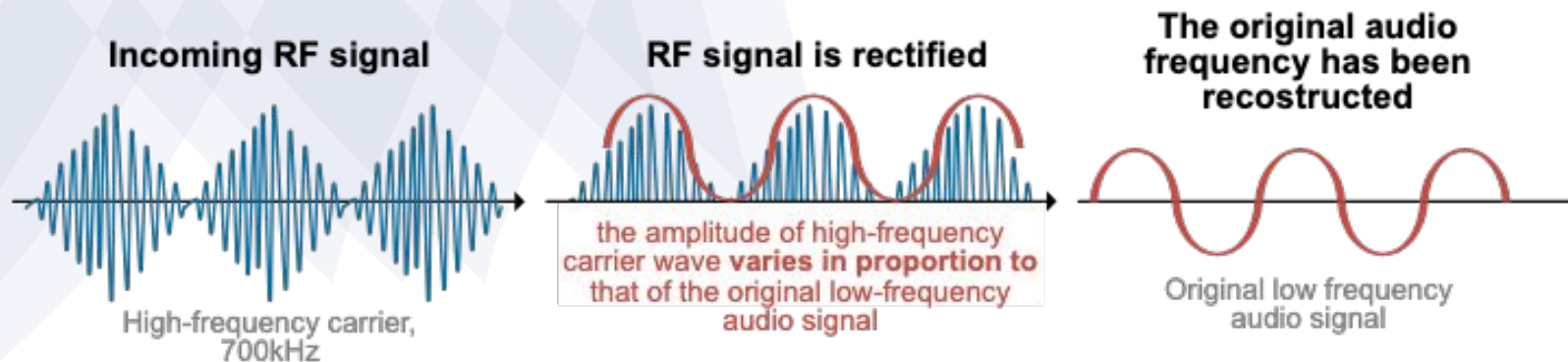
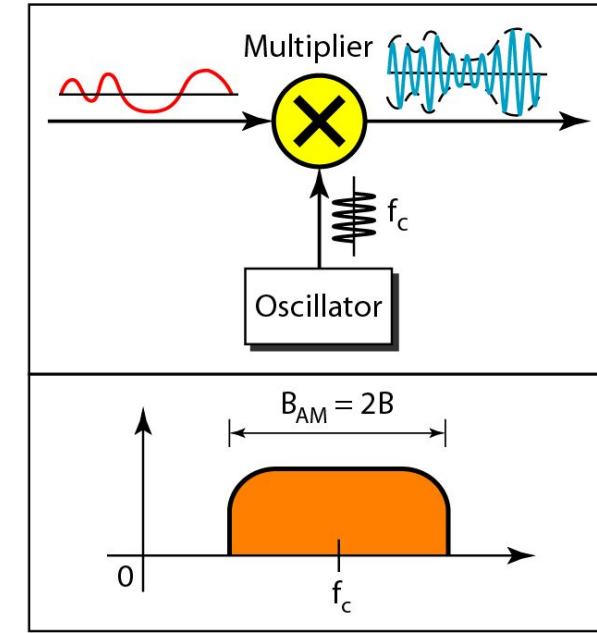
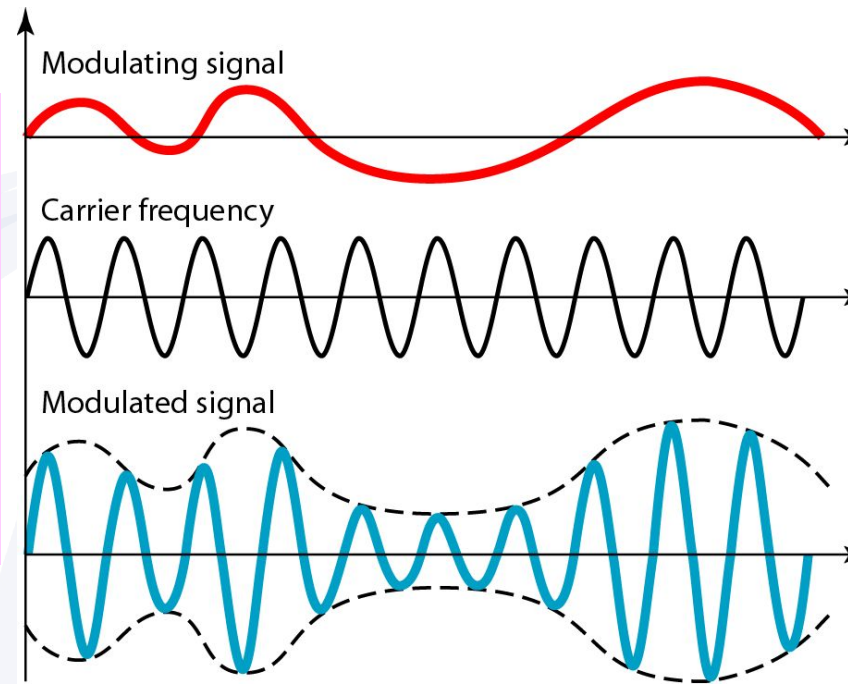


Amplitude Modulation (AM)

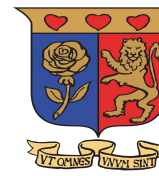


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- ✓ The **frequency** and **phase** of the carrier remain the same;
- ✓ only the **amplitude changes** to follow variations in the information.
- ✓ The **modulating signal** is the **envelope** of the carrier

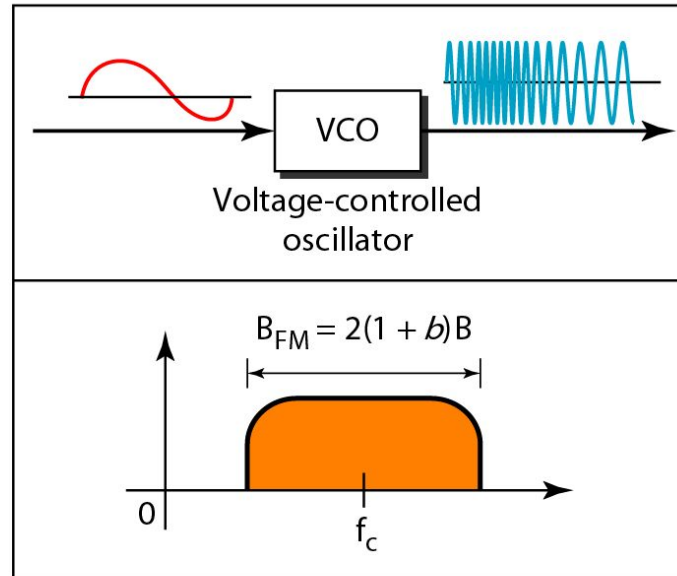
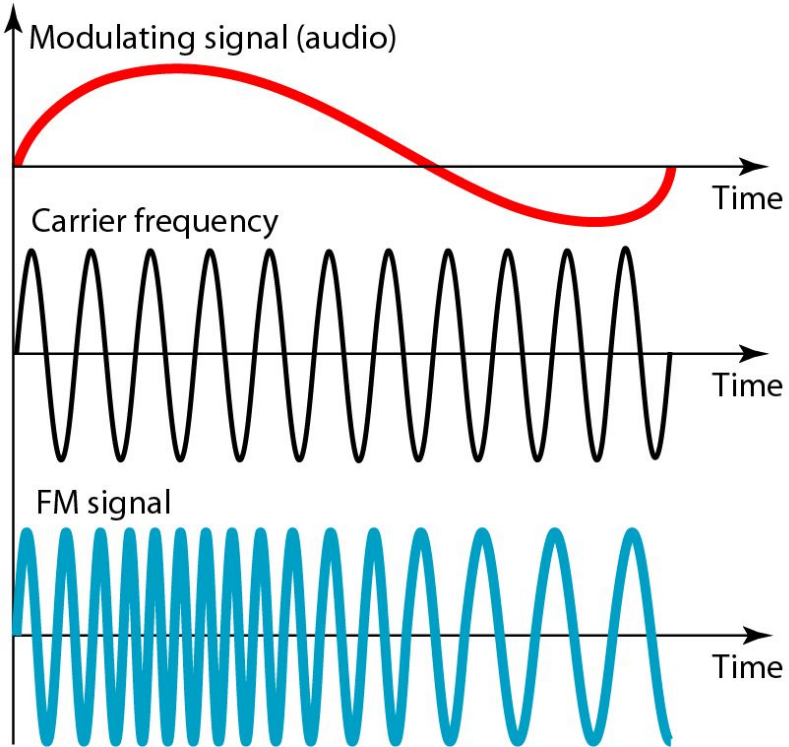


Frequency Modulation (FM)

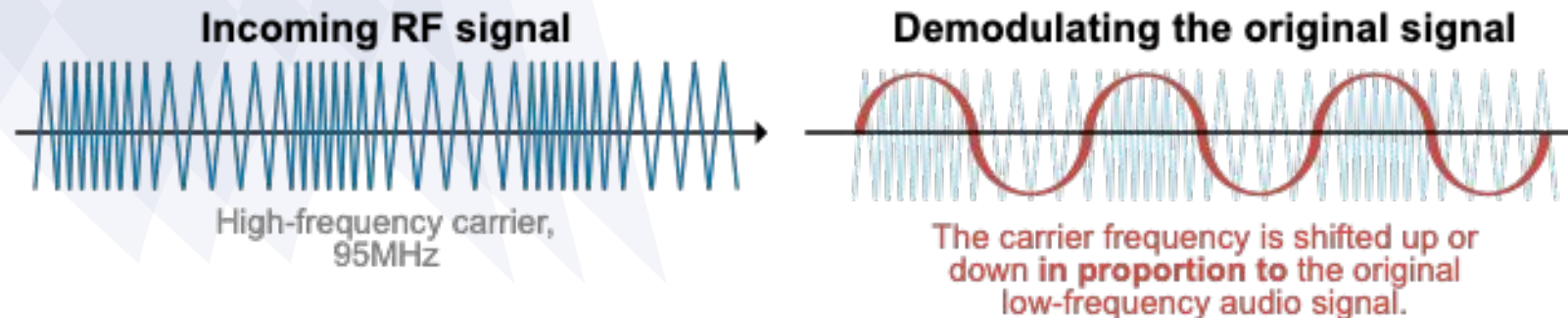


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Amplitude

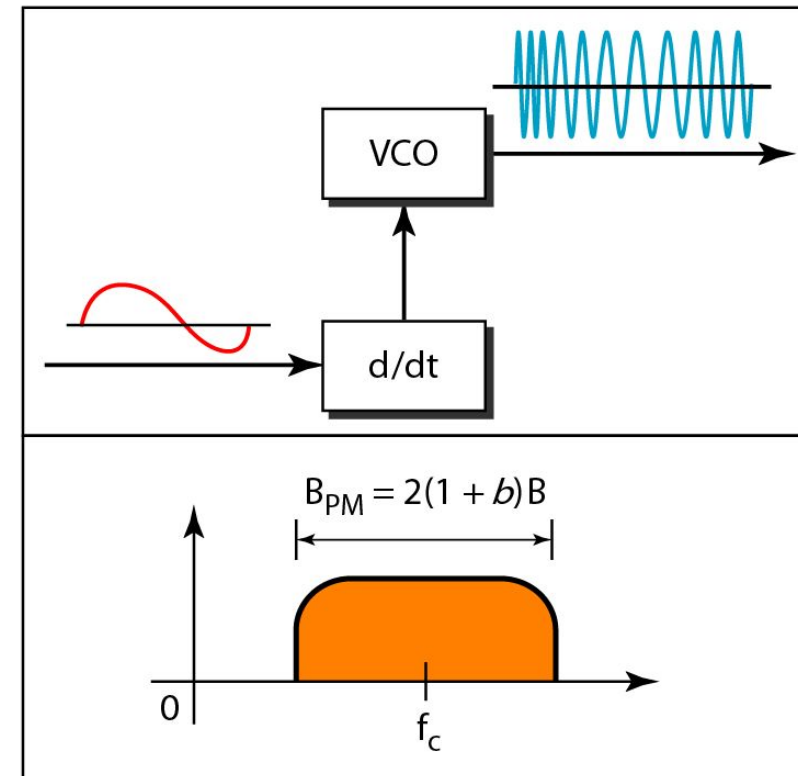
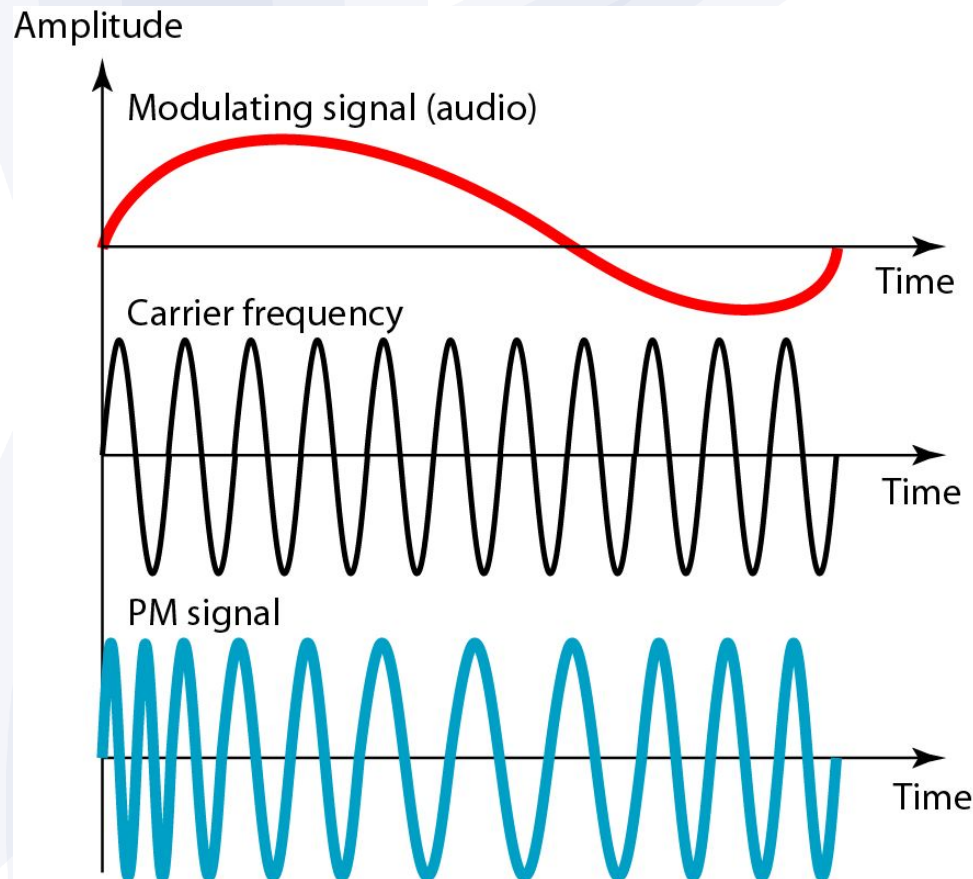


- ✓ The modulating signal changes the frequency f_c of the carrier signal
- ✓ The bandwidth for FM is high
 - It is approx. 10x the signal frequency

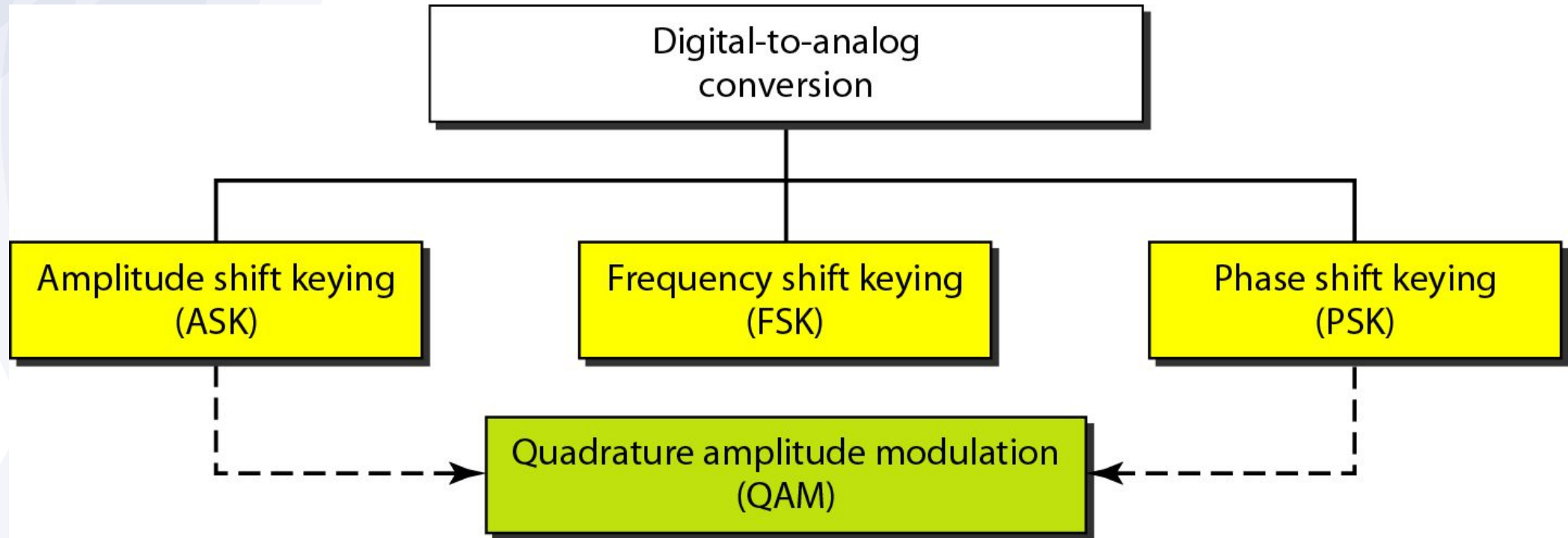
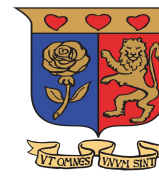


Phase Modulation (PM)

- ✓ The **modulating signal** only changes the **phase** of the **carrier signal**.
- ✓ The **bandwidth** is **higher** than for **AM**.
- ✓ It is used in some systems as an alternative to frequency modulation.

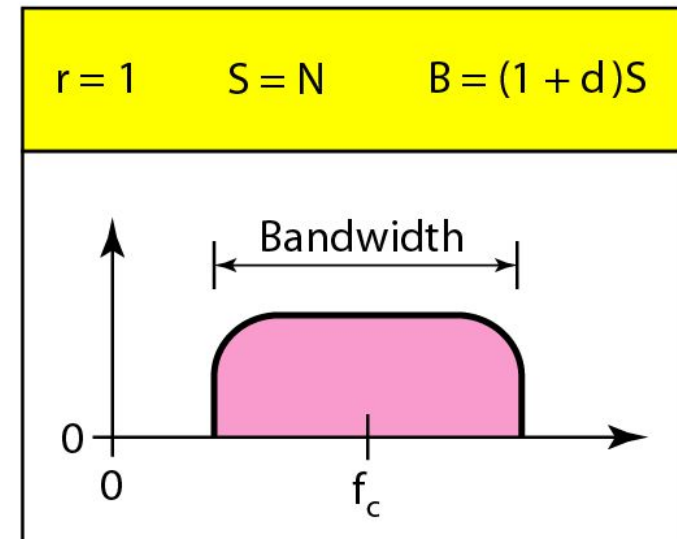
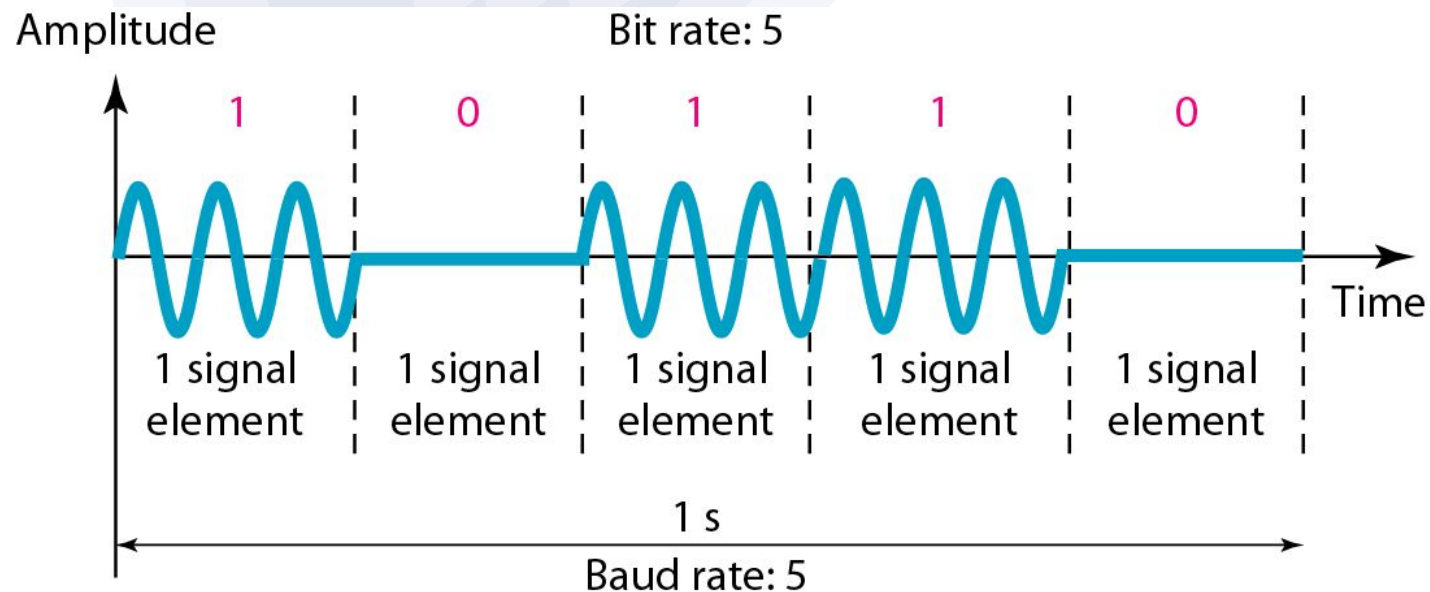


Types of Digital-to-Analog Conversion

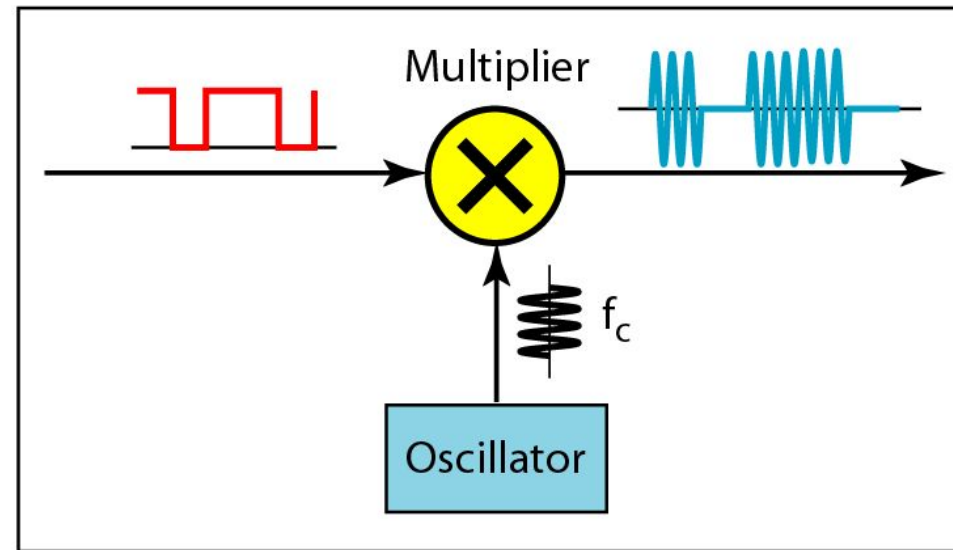
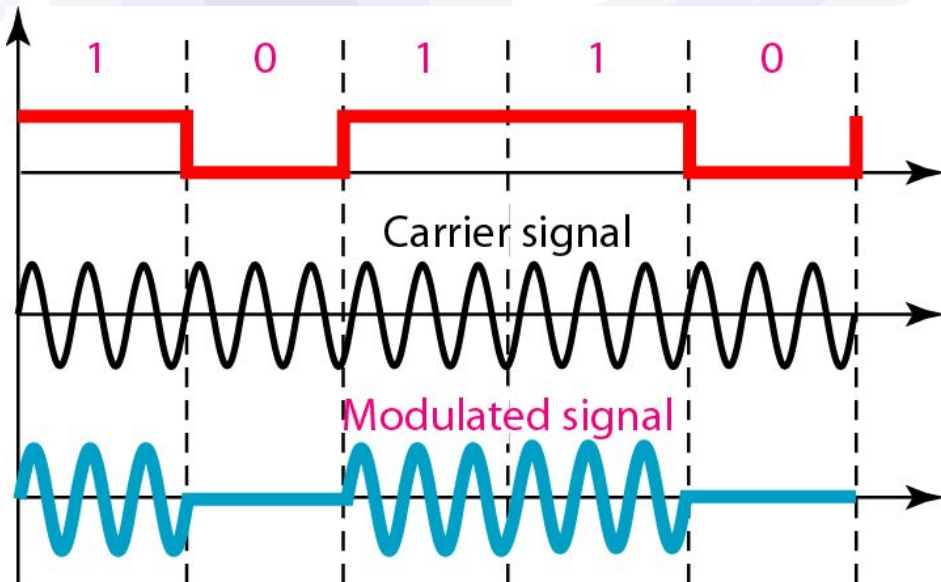


Amplitude Shift Keying (ASK)

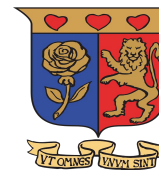
- ✓ The **amplitude** of the **carrier signal** is **varied** to **create signal element**.
- ✓ Both frequency and phase remain constant while the amplitude changes.
- ✓ Highly susceptible to noise interference



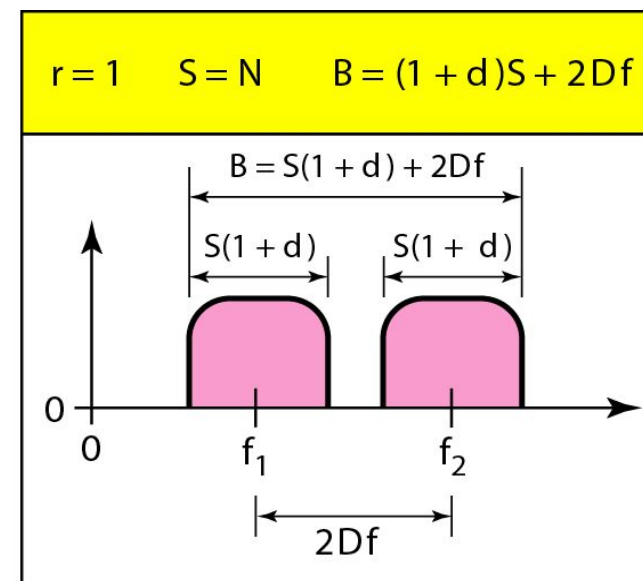
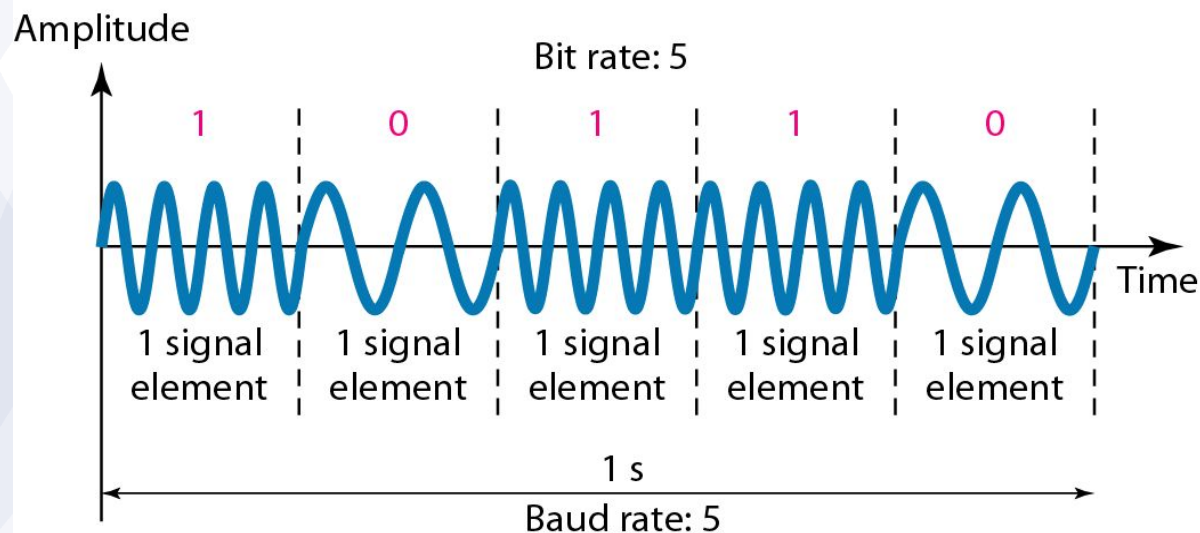
Implementation of binary ASK

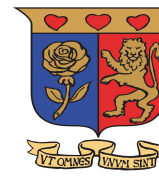


Frequency Shift Keying (FSK)

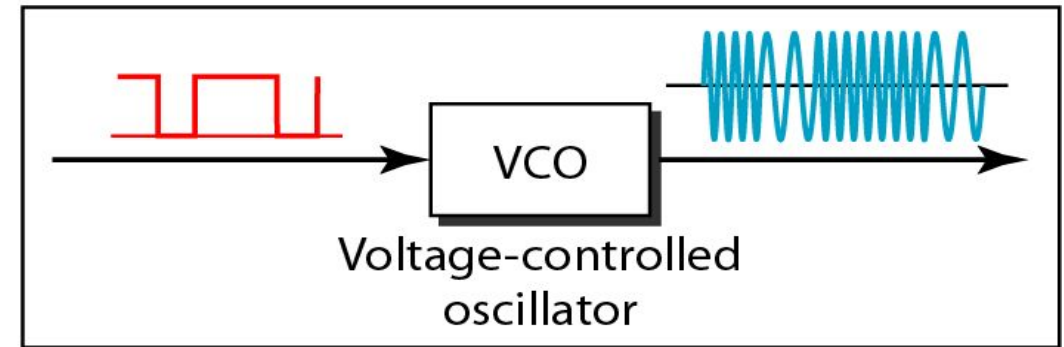
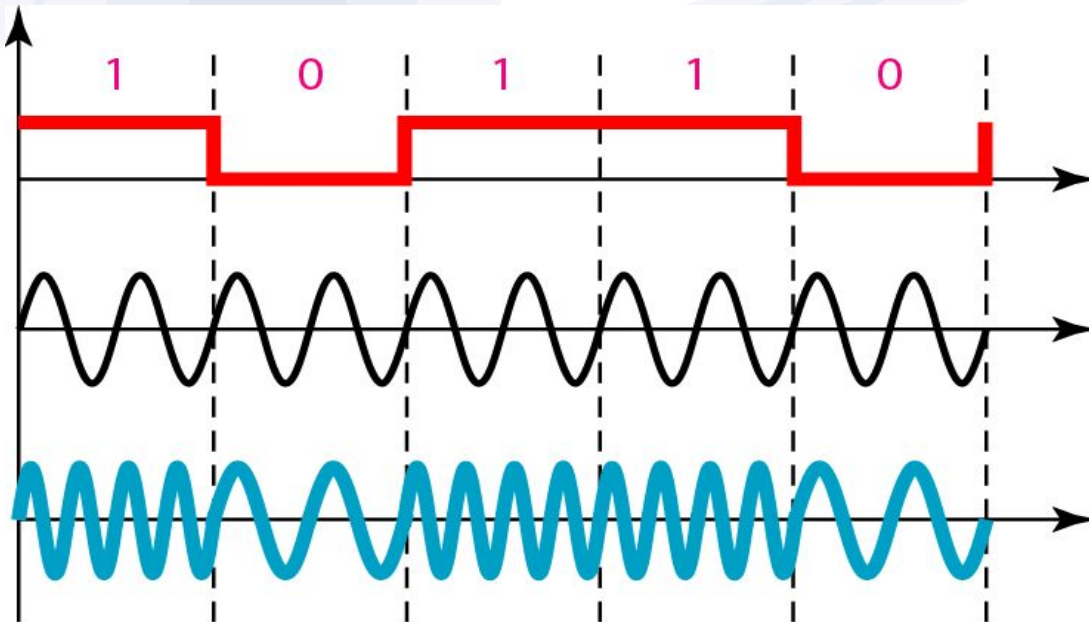


- ✓ The frequency of the carrier signal is varied to represent binary 1 or 0. (Peak amplitude and phase remain constant)
- ✓ For example, a “1” could be represented by $f_1 = f_c + \Delta f$, and a “0” could be represented by $f_2 = f_c - \Delta f$.
- ✓ Both f_1 & f_2 are Δf apart from the midpoint between the two bands.
- ✓ The difference between the two frequency is $2\Delta f$

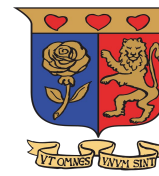




Implementation of binary FSK



Multilevel FSK (MFSK)

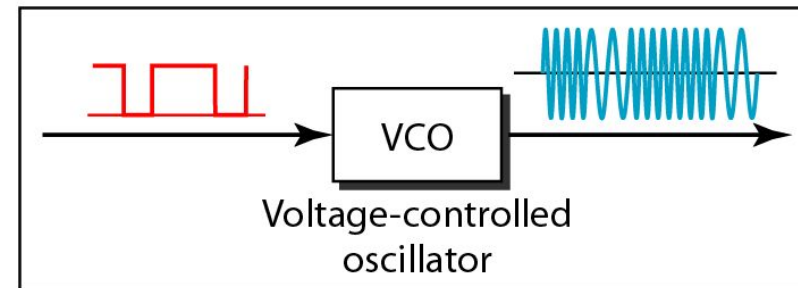
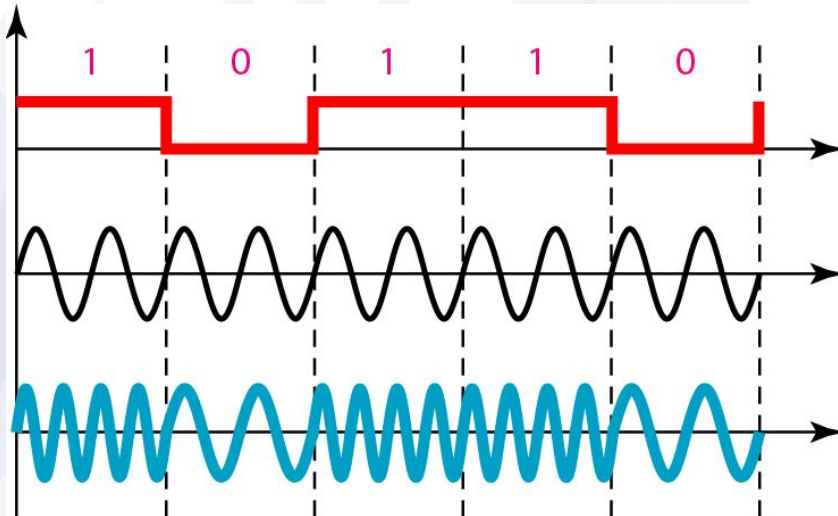


- ✓ We can send data 2-bits at a time by using 4 frequencies.

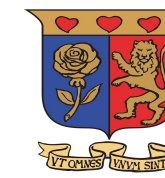
$$B_{\text{MFSK}} (\text{Bandwidth}) = (1+d) \times S + (L-1)2\Delta f \Rightarrow L \times S$$

- The minimum value of $2\Delta f$ should be at least S for the proper operation of modulation and demodulation.

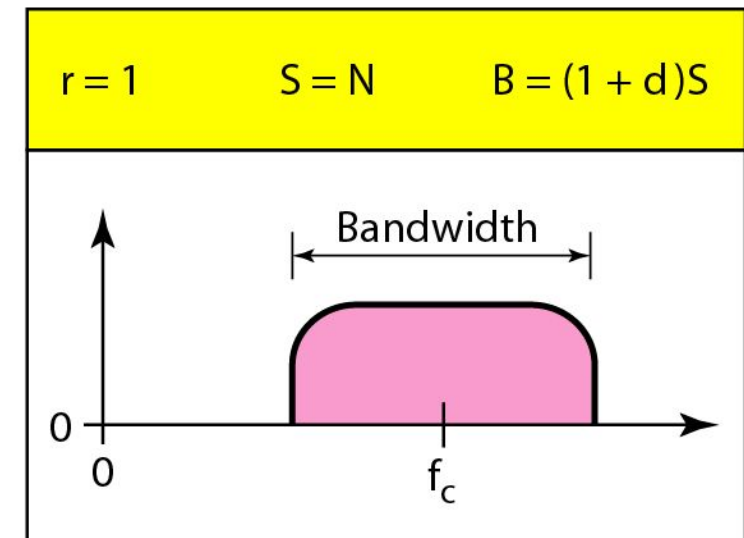
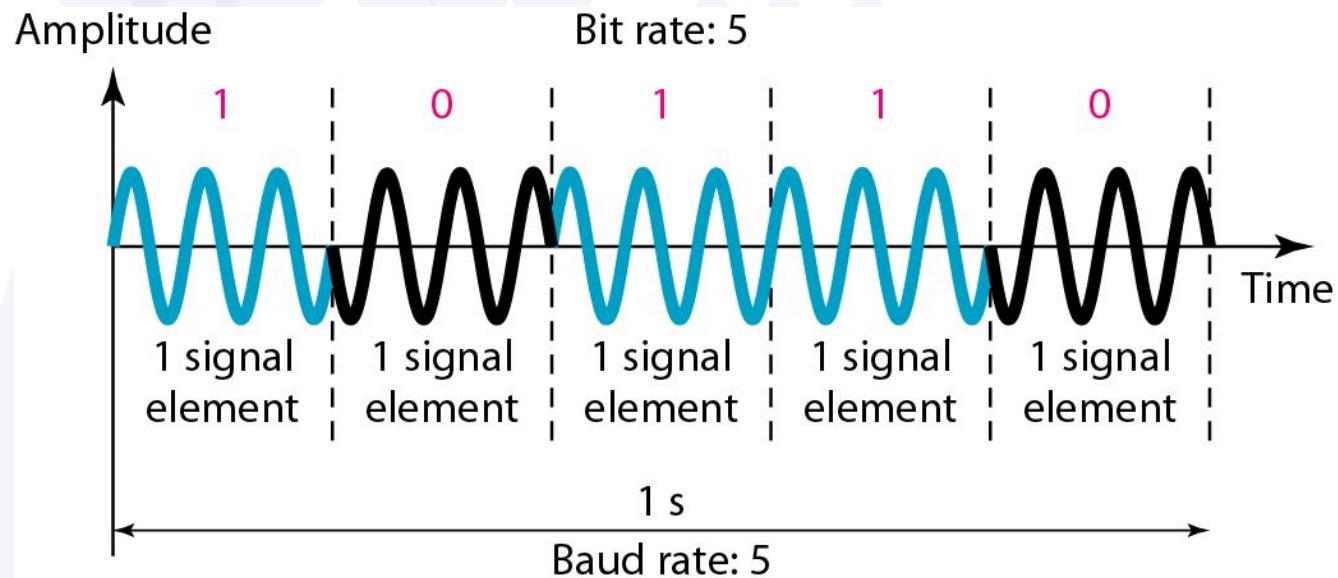
Here, we need to provision for multiple frequencies, each one to represent a group of data bits.



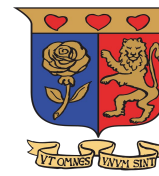
Phase Shift Keying (PSK)



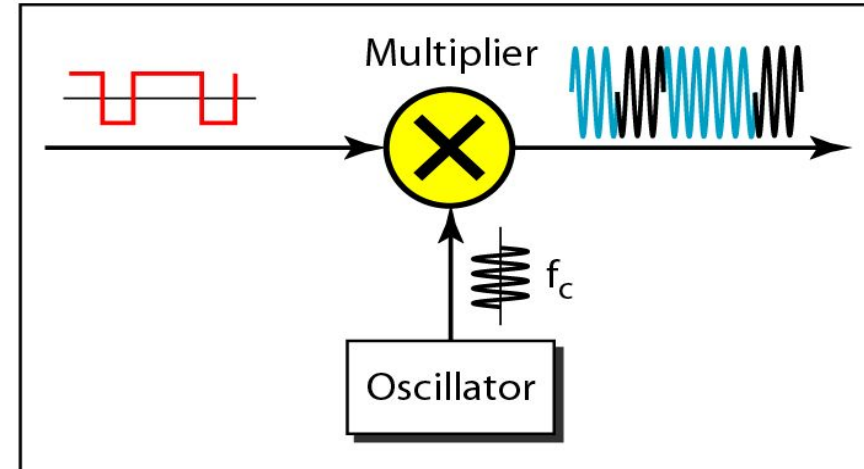
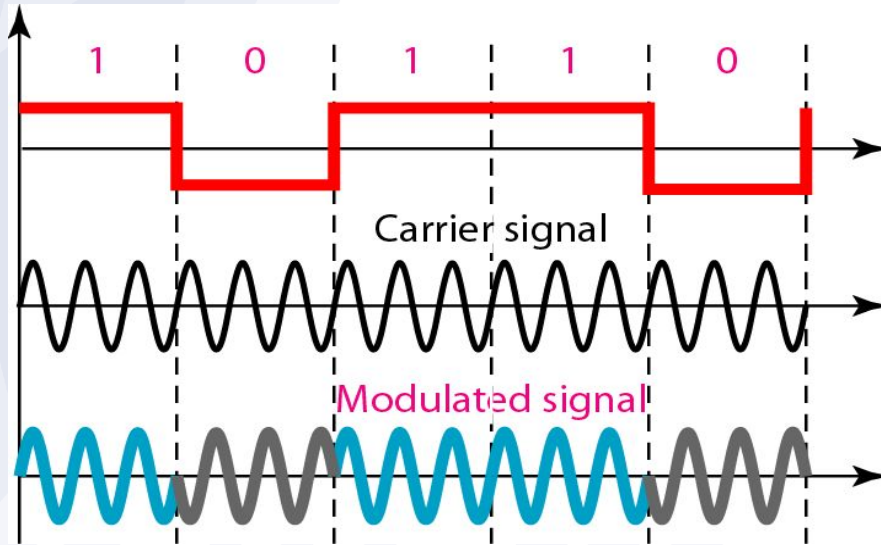
- ✓ The **phase** of the **carrier** is **varied** to represent **two or more different signal elements**.
- ✓ Both peak amplitude and frequency remain constant as the phase changes.
- ✓ PSK is much more robust than ASK as it is not that vulnerable to noise, which changes amplitude of the signal.



Implementation of binary PSK

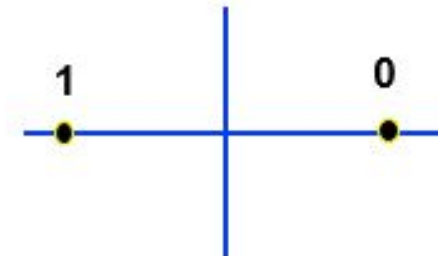


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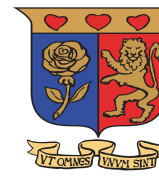
□ The phase is varied to represent binary 1 or 0.

bit	phase
0	0°
1	180°

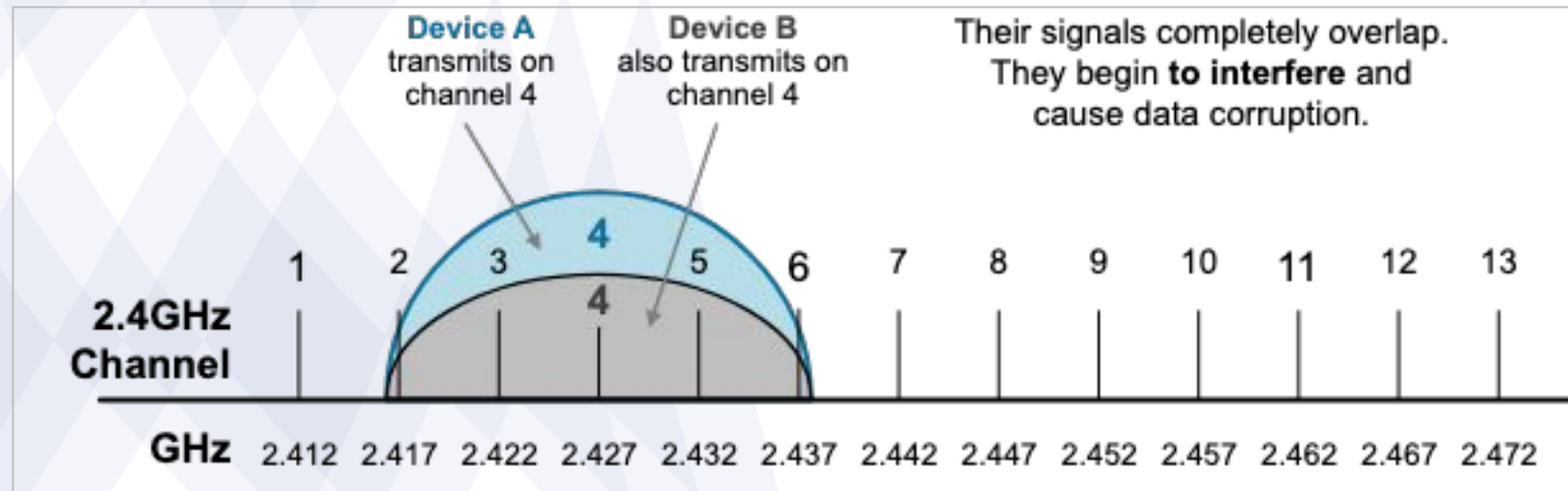


Constellation diagram

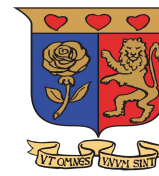
Wi-Fi Interference: Co-channel Interference



- Interference in wireless communications occurs when unwanted signals disrupt the transmission and reception of data. Think of it like trying to have a conversation in a noisy room—the noise makes it hard to hear and understand each other.
- Co-channel interference (CCI) occurs when multiple wireless devices use the same Wi-Fi channel in the same place. For example, devices A and B transmit on the same channel 4 in the 2.4 GHz band. What happens is that their signals completely overlap throughout the entire 22-Mhz width of the channel, as shown in the diagram below.

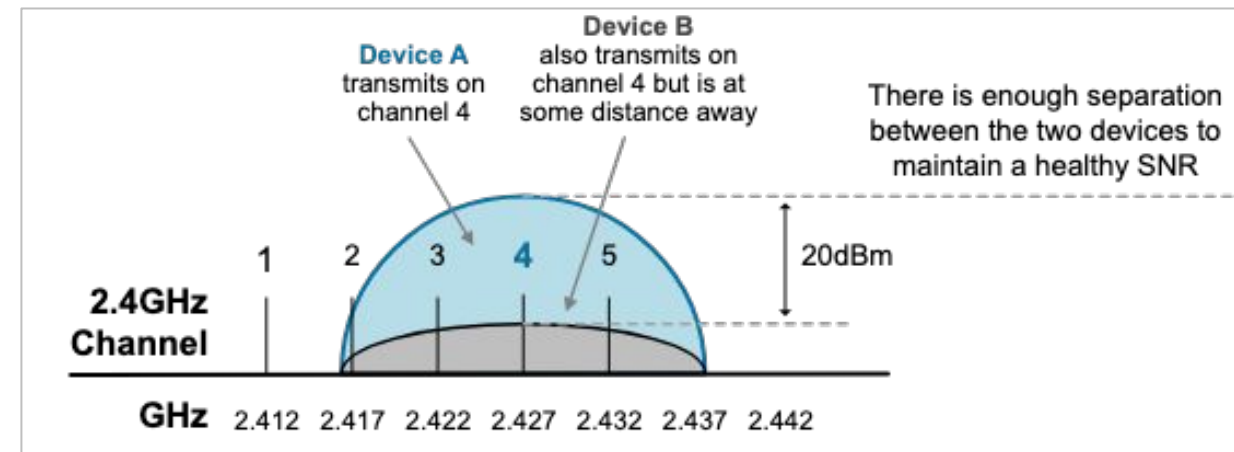
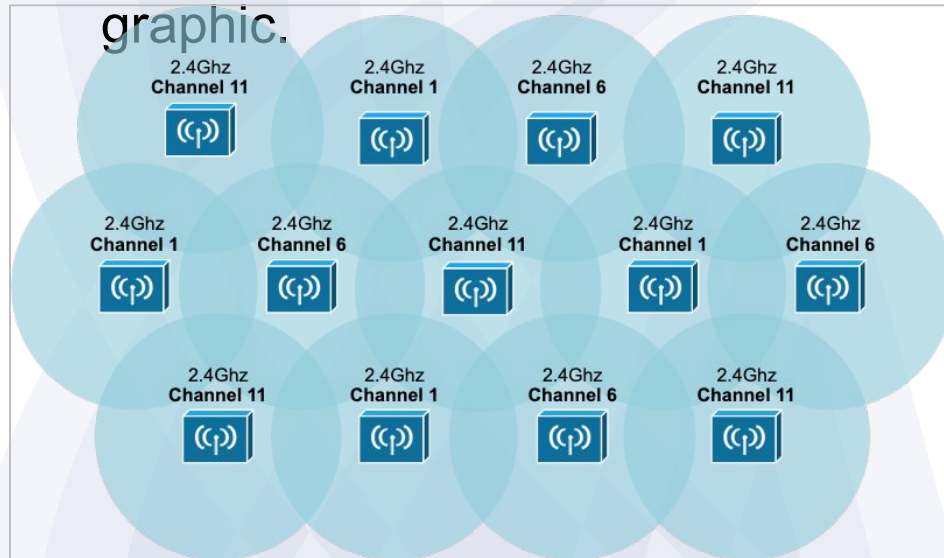


Wi-Fi Interference: Co-channel Interference



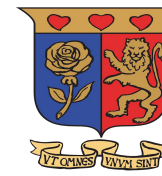
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- Although co-channel interference is obviously harmful, it is unavoidable in most wireless deployments. Both the 2.4GHz and 5GHz bands have only a limited number of channels. Depending on the scale of the wireless LAN and the number of access points (AP), it may be impossible to use non-overlapping channels everywhere. Reducing the effects of co-channel interference always starts with the proper network design, as shown in the first graphic.

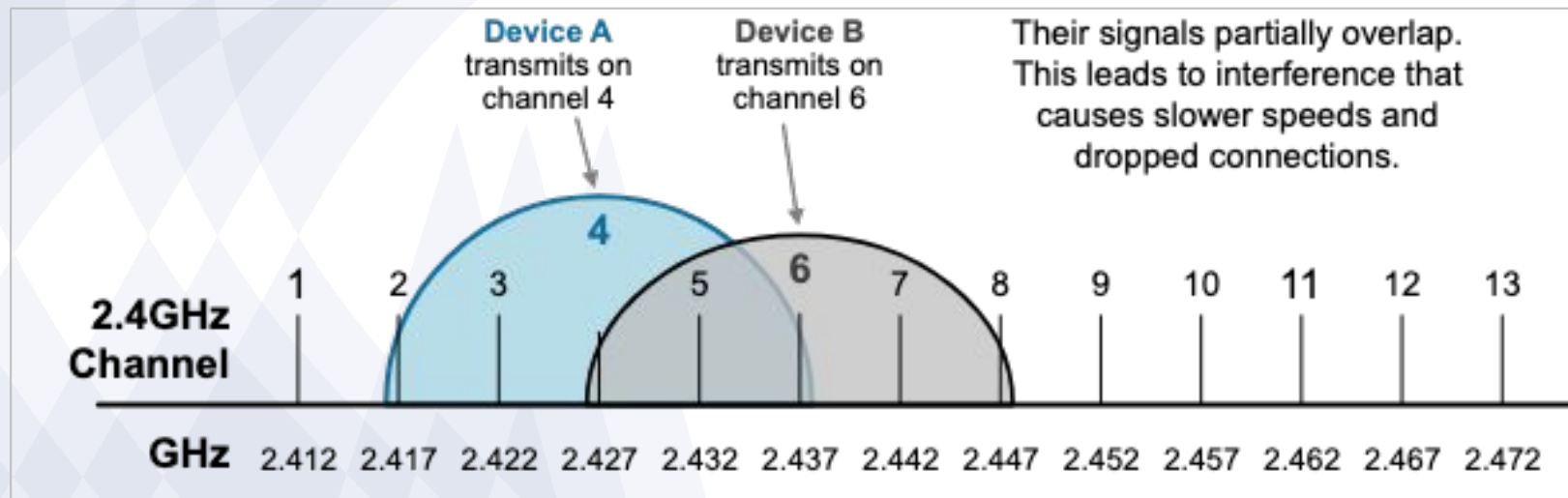


- A good design takes advantage of wall attenuation. You configure an AP to use the same channel as another AP only if they are at least two walls away from each other. This provides enough separation so that the co-channel interference (CCI) is insignificant and both access points can maintain a healthy signal-to-noise ratio (SNR) in relation to each

Wi-Fi Interference: Neighbouring Channel Interference

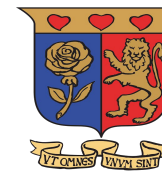


- Neighbouring Channel Interference (NCI) happens when two wireless signals on nearby frequencies interfere with each other. This usually occurs when Wi-Fi devices operate on channels that are too close together. Recall that a WiFi channel has a bandwidth of 22 MHz.
- This means that if a device transmits on channel 4, for example, it uses frequencies between 2.416 and 2.438 GHz and not only the central channel frequency of 2.427 GHz. Therefore, channels can interfere with each other, as shown in the diagram below..



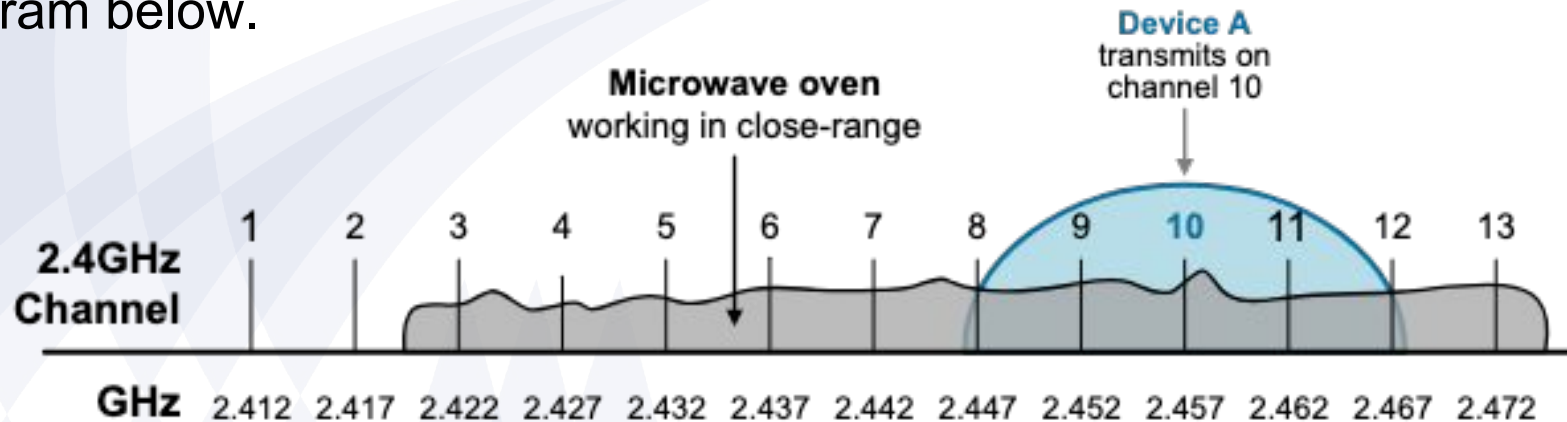
- To avoid NCI, it's best to use non-overlapping channels. The best channels to use in the 2.4 GHz Wi-Fi band are 1, 6, and 11 because they do not overlap. More channels are available in the 5 GHz band, providing more flexibility in designing the wireless LAN so that NCI does not occur. However, a good design may require more access points (AP) configured to transmit at lower power levels.

Non-WiFi interference



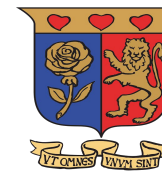
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- In the real world, even non-WiFi devices can cause interference on a particular channel or even the entire Wi-Fi band. For example, a microwave oven near an access point can cause signal disruption, slower speeds, and connection drops on all 2.4GHz channels, as shown in the diagram below.



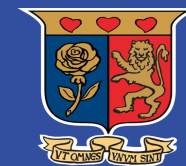
- There could be many sources of non-Wi-Fi interference in every frequency band (2.4GHz, 5GHz, and 6GHz). The most common ones include: Microwaves, Bluetooth devices, cellular phones, Baby monitors, Wireless cameras, Radar systems (especially in 5 GHz) etc.
- This type of interference is more challenging to detect and avoid. In some situations when the 2.4GHz band has a lot of non-Wi-Fi signals, which causes the Noise Floor to be very high, it may be better to reconfigure the entire wireless LAN to another band (for instance, 5GHz) rather than trying to design around the external noise

Free Space Path Loss and Dynamic Rate Shifting



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- When a radio frequency (RF) signal is transmitted from an antenna, its strength decreases as it moves through open space. Even with no obstacles in the way, the signal still weakens. This effect is called free space path loss (FSPL)—even signals traveling through the vacuum of space experience FSPL.
- RF signals don't move in a straight line; instead, they spread out in waves. Imagine the signal forming a sphere that expands as it travels. The energy remains the same, but as the sphere grows, the signal gets weaker because it's spread over a larger area. This happens no matter what kind of antenna is used, even if it focuses energy in a tight beam.
- The loss is relative to two RF signal components: Frequency and Distance.
- Wi-Fi devices automatically adjust their modulation and coding based on signal strength. If the signal is strong, they use complex methods to send more data at higher speeds. If the signal weakens, they switch to simpler methods to maintain a stable connection, even if it means slower speeds. This process is called *dynamic rate shifting (DRS)* and happens automatically without user intervention. Different manufacturers may use slightly different approaches, but the goal is the same—adjusting data transmission to match real-time signal conditions.
- In summary, free space path loss is unavoidable, but Wi-Fi devices are designed to adapt and ensure stable communication over different distances and environments.



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